1 The Foundations: Logic and Proofs

- **1.1** Propositional Logic
- 1.2Applications of Propositional Logic
  1.3 Propositional Equivalences
  1.4 Predicates and Quantifiers
  1.5 Nested Quantifiers
  1.6 Rules of Inference
  1.7 Introduction to Proofs
  1.8 Proof Methods and Strategy



Foundations of Logic

Rosen 8th ed., §§1.1-1.5

数理逻辑(symbolic logic)

逻辑：是研究推理的科学。公元前四世纪由希腊的亚里斯多德首创。作为一门独立科学，十七世纪，德国莱布尼兹给逻辑学引进了符号，又称为数理逻辑。

利用计算的方法来代替人们思维中的逻辑推理过程，这种想法早在十七世纪就有人提出过。莱布尼茨就曾经设想过能不能创造一种“通用的科学语言”，可以把推理过程象数学一样利用公式来进行计算，从而得出正确的结论。由于当时的社会条件，他的想法并没有实现。但是他的思想却是现代数理逻辑部分内容的萌芽，从这个意义上讲，莱布尼茨可以说是数理逻辑的先驱 。

1847年，英国数学家布尔发表了《逻辑的数学分析》，建立了“布尔代数”，并创造一套符号系统，利用符号来表示逻辑中的各种概念。布尔建立了一系列的运算法则，利用代数的方法研究逻辑问题，初步奠定了数理逻辑的基础。

十九世纪末二十世纪初，数理逻辑有了比较大的发展，**1884**年，德国数学家弗雷格出版了《算术基础》一书，在书中引入量词的符号，使得数理逻辑的符号系统更加完备。对建立这门学科做出贡献的，还有美国人皮尔斯，他也在著作中引入了逻辑符号。从而使现代数理逻辑最基本的理论基础逐步形成，成为一门独立的学科。

Foundations of Logic: Overview

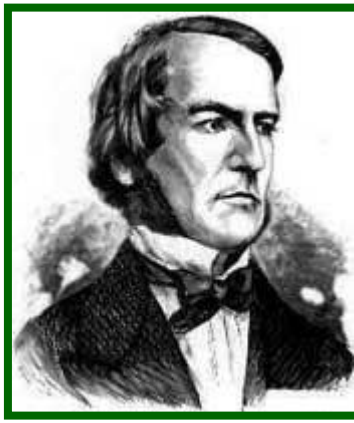
- Propositional logic (§1.1-1.3): 命题逻辑
  - Basic definitions. (§1.1)
  - Applications (§1.2)
  - Equivalence rules (§1.3)
- Predicate logic (§1.4-1.5) 谓词逻辑
  - Predicates.
  - Quantified predicate expressions.
  - Equivalences

Propositional Logic (§ 1.1)

Propositional Logic is the logic of compound statements built from simpler statements using so-called *Boolean connectives.*

Some applications in computer science:

- Design of digital electronic circuits.
- Expressing conditions in programs.
- Queries to databases & search engines.



George Boole
(1815-1864)

Propositions in natural language

Definition: A *proposition* is simply:

- a *statement* (*i.e.*, a declarative sentence)
  - *with some definite meaning*, (not vague or ambiguous)
- having a *truth value* that's either *true* (**T**) or *false* (**F**)
  - it is **never** both, neither, or somewhere "in between!"
    - However, you might not *know* the actual truth value
- Later, we will study *probability theory,* in which we assign *degrees of certainty* ("between" **T** and **F**) to propositions.
  - But for now: think True/False only!

Propositions（命题）

- A **statement** or **proposition** is a declarative sentence（陈述句） that is either *true* or *false*,but not both.
  - true = T (or 1)
  - false = F (or 0) (binary logic)

Example

- ‘The moon is made of green cheese’
- ‘go to town!’
- X - *imperative*
- ‘What time is it?’
- X – *interrogative*

Examples of NL Propositions

- "It is raining." (In a given situation.)
- "Beijing is the capital of China, and 1 + 2 = 2"

But, the following are **NOT** propositions:

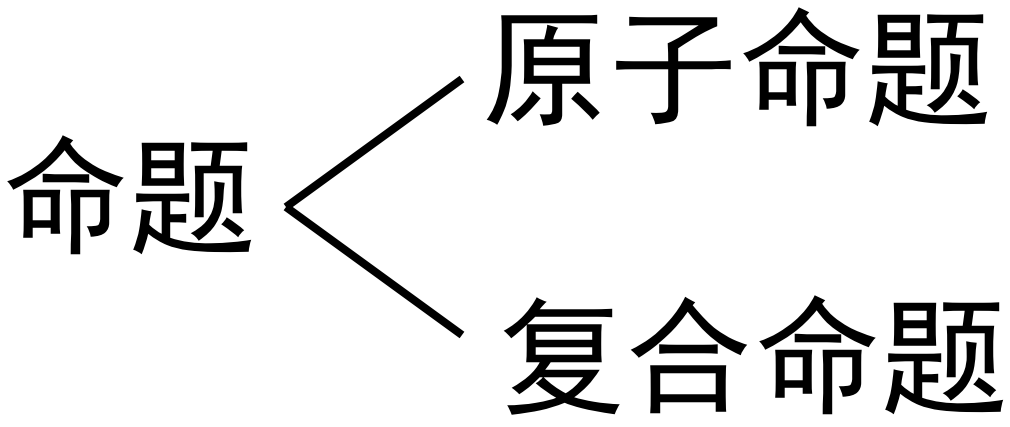
- "La la la la la." (no meaning)
- "Who's there?" (interrogative: no truth value)
- "x := x+1" (imperative: no truth value)
- "1 + 2" (term: no truth value)

Example 1

- Which of the following are propositions?
  - (a) The earth is round.
  - (b) 2+3=5
  - (c) Do you speak English?
  - (d) 3-x=5
  - (e) Take two aspirins.
  - (f) The temperature on the surface of the planet Venus is 800ºF.
  - (g) The sun will come out tomorrow.

Propositional variables

- We use letters to denote **propositional variables** (or statement variables), that is, variables that represent propositions, just as letters are used to denote numerical variables. The conventional letters used for propositional variables are p, q, r, s, . . .
- The area of logic that deals with propositions is called the **propositional calculus** or **propositional logic**.



原子命题：不能分解为更简单的陈述句。

Propositions in Propositional Logic

- **Atoms**: p, q, r, …
  (Corresponds with simple English sentences, e.g. `I had salad for lunch')

- **Compound propositions** : built up from atoms using operators: $p \wedge q$
  (Corresponds with compound English sentences, e.g., "I had salad for lunch **and** I had steak for dinner.")

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义务教育教科书

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数学

一年级 上册

主 编 刘 坚 孔企平 张 丹
本册主编 刘可钦 张 红



·北京·

亲爱的小朋友：



智慧老人

你知道“数学”是什么样子的？

“数学”是数字，学了数字可以看时间、看日期。





淘气

我认为“数学”是页码，是24支彩笔，是各种各样的形状……



笑笑





江江，这里面有数学！



机灵狗

小朋友，数学就在我们身边，你和数学一定会成为好朋友。

编者大朋友

Operators / Connectives

An *operator* or *connective* combines *n* *operand* expressions into a larger expression. (*E.g.*, "+" in numeric exprs.)

- *Unary* operators take 1 operand (*e.g.*, −3); *binary* operators take 2 operands (*eg* 3×4).
- *Propositional* or *Boolean* operators operate on propositions instead of on numbers.

Logical Operators(逻辑运算）

- Unary
  - **Negation**（否定）
- Binary
  - **Conjunction**（合取）
  - **Disjunction** （析取）
  - **Exclusive OR**（异或）
  - **Implication** （蕴涵）
  - **Biconditional** （等价）

Some Popular Boolean Operators

Formal Name	**Nickname**	**Arity**	**Symbol**
Negation operator	NOT	Unary	$\neg$
Conjunction operator	AND	Binary	$\wedge$
Disjunction operator	OR	Binary	$\vee$
Exclusive-OR operator	XOR	Binary	$\oplus$
Implication operator	IMPLIES	Binary	$\rightarrow$
Biconditional operator	IFF	Binary	$\leftrightarrow$

加法口诀表



1	2	3	4	5	6	7	8	9
1+1=2								
1+2=3	2+1=3							
1+3=4	2+2=4	3+1=4						
1+4=5	2+3=5	3+2=5	4+1=5					
1+5=6	2+4=6	3+3=6	4+2=6	5+1=6				
1+6=7	2+5=7	3+4=7	4+3=7	5+2=7	6+1=7			
1+7=8	2+6=8	3+5=8	4+4=8	5+3=8	6+2=8	7+1=8		
1+8=9	2+7=9	3+6=9	4+5=9	5+4=9	6+3=9	7+2=9	8+1=9	
1+9=10	2+8=10	3+7=10	4+6=10	5+5=10	6+4=10	7+3=10	8+2=10	9+1=10

减法口诀表



10	9	8	7	6	5	4	3	2
10-1=9								
10-2=8	9-1=8							
10-3=7	9-2=7	8-1=7						
10-4=6	9-3=6	8-2=6	7-1=6					
10-5=5	9-4=5	8-3=5	7-2=5	6-1=5				
10-6=4	9-5=4	8-4=4	7-3=4	6-2=4	5-1=4			
10-7=3	9-6=3	8-5=3	7-4=3	6-3=3	5-2=3	4-1=3		
10-8=2	9-7=2	8-6=2	7-5=2	6-4=2	5-3=2	4-2=2	3-1=2	
10-9=1	9-8=1	8-7=1	7-6=1	6-5=1	5-4=1	4-3=1	3-2=1	2-1=1

昵图网 www.nipic.com By:1242150593 No.20131112155918302143

The Negation Operator

The unary *negation operator* "¬" (*NOT*) transforms a prop. into its *negation*.

E.g. If p = "I have brown hair."

then $\neg p$ = "I do **not** have brown hair."

The *truth table* 真值表for NOT:

p	$\neg p$
T	F
F	T

Operand column

Result column

T :≡ True; F :≡ False

":≡" means "is defined as"

- 1、命题(Proposition)
- 2、从简单命题(atomic proposition)到
- 复合命题(compositional proposition)
- 3、从命题常量(propositional constant)到
- 命题变量(propositional variable)
- 4、从复合命题(compositional proposition)到
- 命题公式(propositional formulas)

真值表TRUTH TABLES

- 在命题公式中，对于分量指派真值的各种可能组合，就确定了这个命题公式的各种真值情况，把它汇列成表，就是命题公式的真值表。

The Conjunction合取 Operator

The binary *conjunction operator* "$\wedge$" (*AND*) combines two propositions to form their logical *conjunction*.

E.g. If *p*="I will have salad for lunch." and *q*="I will have steak for dinner.", then

$p \wedge q$="I will have salad for lunch **and** I will have steak for dinner."

Conjunction Truth Table

- Note that a conjunction $p_1 \wedge p_2 \wedge \ldots \wedge p_n$ of n propositions will have 2^n rows in its truth table.

Operand columns

p	q	$p \wedge q$
F	F	F
F	T	F
T	F	F
T	T	T

- Also: $\neg$ and $\wedge$ operations together are suffi-cient to express *any* Boolean truth table!

The Disjunction析取 Operator

The binary *disjunction operator* "$\vee$" (*OR*) combines two propositions to form their logical *disjunction*.

p="My car has a bad engine."

q="My car has a bad door."

$p \vee q$="Either my car has a bad engine, **or** my car has a bad door."

Meaning is like "and/or" in English.

Disjunction Truth Table

- Note that $p \vee q$ means that *p* is true, or *q* is true, **or both** are true!
- So, this operation is also called *inclusive or,* because it **includes** the possibility that both *p* and *q* are true.
- "$\neg$" and "$\vee$" together are also universal.

p	q	$p \vee q$
F	F	F
F	T	**T**
T	F	**T**
T	T	T

Note difference from AND

Let’s introduce some additional connectives

- A variant of disjunction
- The conditional
- The biconditional

The *Exclusive Or* Operator

The binary *exclusive-or operator* "⊕" (*XOR*) combines two propositions to form their logical "exclusive or".

p = "I will earn an A in this course,"

q = "I will drop this course,"

$p \oplus q$ = "I will either earn an A in this course, or I will drop it (but not both!)"

Exclusive-Or Truth Table

- Note that $p \oplus q$ means that p is true, or q is true, but **not both**!
- This operation is called *exclusive or,* because it **excludes** the possibility that both p and q are true.
- "¬" and "⊕" together are **not** universal.

p	q	$p \oplus q$
F	F	F
F	T	T
T	F	T
T	T	**F**

Note difference from OR.

Natural Language is Ambiguous

Note that English "or" can be ambiguous regarding the "both" case!

p	q	p "or" q
F	F	F
F	T	T
T	F	T
T	T	?

Need context to disambiguate the meaning!

For this class, assume "or" means inclusive

The *Implication* Operator

antecedent consequent

The *implication* $p \rightarrow q$ states that *p* implies *q.*

I.e., If *p* is true, then *q* is true; but if *p* is not true, then *q* could be either true or false.

E.g., let *p* = "You study hard."
q = "You will get a good grade."

$p \rightarrow q$ = "If you study hard, then you will get a good grade."

Implication Truth Table

- $p \rightarrow q$ is **false** only when p is true but q is **not** true.
- $p \rightarrow q$ does **not** say that p causes q!
- $p \rightarrow q$ does **not** require that p or q **are true**!
- *E.g.* "(1=0) $\rightarrow$ pigs can fly" is TRUE!

p	q	$p \rightarrow q$
F	F	T
F	T	T
T	F	**F**
T	T	T

The only False case!

《大 学》

- 大学之道，在明明德，在亲民，在止于至善。知止而后有定；定而后能静；静而后能安；安而后能虑；虑而后能得。物有本末，事有终始。知所先后，则近道矣。
- 古之欲明明德于天下者，先治其国；欲治其国者，先齐其家；欲齐其家者，先修其身；欲修其身者，先正其心；欲正其心者，先诚其意；欲诚其意者，先致其知；致知在格物。物格而后知至；知至而后意诚；意诚而后心正；心正而后身修；身修而后家齐；家齐而后国治；国治而后天下平。

Different Ways of Expressing $p \rightarrow q$

- **if** p, **then** q
- **if** p, q
- q **unless** $\neg p$
- q **if** p
- q **whenever** p
- q **follows from** p

- p **implies** q
- p **only if** q
- q **when** p
- q **when** p
- p **is sufficient for** q
- q **is necessary for** p

- **a necessary condition for** p **is** q
- **a sufficient condition for** q **is** p

English Phrases Meaning $p \rightarrow q$

- "*p* implies *q*"
- "if *p*, then *q*"
- "if *p*, *q*"
- "when *p*, *q*"
- "whenever *p*, *q*"
- "*q* if *p*"
- "*q* when *p*"
- "*q* whenever *p*"
- "*p* only if *q*"
- "*p* is sufficient for *q*"
- "*q* is necessary for *p*"
- "*q* follows from *p*"
- "*q* is implied by *p*"
- "*q* unless ¬ *p*"

- “*q* unless ¬ *p*”
- Unless you finish your homework, I will not take you out for play.除非你完成你的家庭作业，不然我不会带你出去玩。
- Unless we listen ,we can’t know how to respond.如果我们不听,就不知道怎么回答。
- Unless he can get a big progress, he will not stay at this position.除非他能取得大的进步，否则他不能待在这个职位上了。
- She will keep on singing unless she is told to stop.她会一直唱下去，除非有人叫她停止。
- Unless you can catch me, I will not marry you.除非你能追上我，不然我不会嫁给你。

Converse, Contrapositive, and Inverse
逆命题 逆否命题 否命题

- From $p \to q$ we can form new conditional statements .
  - $q \to p$ is the **converse** of $p \to q$
  - $\neg q \to \neg p$ is the **contrapositive** of $p \to q$
  - $\neg p \to \neg q$ is the **inverse** of $p \to q$

Example: Find the converse, inverse, and contrapositive of "It raining is a sufficient condition for my staying at home."

Solution:

converse: If I stay at home, then it is raining.

inverse: If it is not raining, then I will not stay at home.

contrapositive: If I do not stay at home, then it is not raining.

Contrapositive

Some terminology, for an implication $p \rightarrow q$:

- Its *converse* is: $q \rightarrow p$.
- Its *contrapositive*: $\neg q \rightarrow \neg p$.
- One of these has the *same meaning* (same truth table) as $p \rightarrow q$. Can you figure out which?

原命题为：p－－>q

converse 逆命题为：q－－>p

inverse 否命题为：非p－－>非q

逆否命题为：非q－－>非p



How do we know for sure?

Proving the equivalence of $p \rightarrow q$ and its contrapositive using truth tables:

p	q	$\neg q$		$\neg p$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
F	F	T	→	T	T	T
F	T	F	→	T	T	T
T	F	T	→	F	F	F
T	T	F	→	F	T	T

The *biconditional* operator

The *biconditional* $p \leftrightarrow q$ states that *p* is true *if and only if (IFF) q* is true.

p = "Joseph Robinette Biden won the *2020* presidential *election*."

q = "Joseph Robinette Biden was president for all of 2021."

$p \leftrightarrow q$ = "If, and only if, Joseph Robinette Biden won the *2020* presidential *election*, Joseph Robinette Biden was president for all of 2021."

Biconditional Truth Table

- $p \leftrightarrow q$ means that p and q have the **same** truth value.
- Note this truth table is the exact **opposite** of $\oplus$'s!

  Thus, $p \leftrightarrow q$ means $\neg(p \oplus q)$
- $p \leftrightarrow q$ does **not** imply that p and q are true, or that either of them causes the other.

p	q	$p \leftrightarrow q$
F	F	T
F	T	F
T	F	F
T	T	T

Boolean Operations Summary

- We have seen 1 unary operator (out of the 4 possible) and 5 binary operators:



p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \rightarrow q$	$p \leftrightarrow q$
F	F	T	F	F	F	T	T
F	T	T	F	T	T	T	F
T	F	F	F	T	T	F	F
T	T	F	T	T	F	T	T

Questions for you to think about

1. Consider a conjunction $p_1 \wedge p_2 \wedge p_3$
   How many rows are there in its truth table?
2. Consider a conjunction
   $p_1 \wedge p_2 \wedge \ldots \wedge p_n$ of n propositions.
   How many rows are there in its truth table?

3. Explain why $\neg$ and $\wedge$ together are sufficient to express *any* Boolean truth table

Questions for you to think about

1. Consider a conjunction $p_1 \wedge p_2 \wedge p_3$
   How many rows are there in its truth table? 8

p_1	p_2	p_3	$p_1 \wedge p_2 \wedge p_3$
1	1	1	
1	1	0	
1	0	1	
1	0	0	
0	1	1	
0	1	0	
0	0	1	
0	0	0	

Questions for you to think about

2. Consider $p_1 \wedge p_2 \wedge \ldots \wedge p_n$
   How many rows are there
   in its truth table?

   2.2.2. … .2 (n factors)
   Hence 2^n (This grows exponentially!)

For you to think about:

1. Can you think of yet another 2-place connective?
   How many possible connectives do there exist?

For you to think about:

1. How many possible connectives do there exist?

p	**q**	**p connective q**
T	T	?
T	F	?
F	T	?
F	F	?

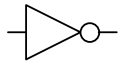
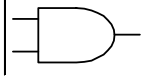
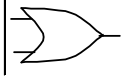
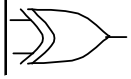
Each question mark can be T or F, hence 2.2.2.2=16 connectives

Example of another connective

p	**connective**	**q**	compare:	**p and q**
T	F	T		T
T	T	F		F
F	T	T		F
F	T	F		F

Names: NAND, Sheffer stroke

Some Alternative Notations

Name:	not	and	or	xor	implies	iff
Propositional logic:	$\neg$	$\wedge$	$\vee$	$\oplus$	$\rightarrow$	$\leftrightarrow$
Boolean algebra:	$\overline{p}$	pq	$+$	$\oplus$		
C/C++/Java (wordwise):	!	&&	||	!=		==
C/C++/Java (bitwise):	~	&	|	^		
Logic gates:						

The language of propositional logic defined more properly

- Atoms: p1, p2, p3, ..
- Formulas:命题公式
  - All atoms are formulas
  - If α is a formula then $\neg \alpha$ is a formula
  - If α and β are formulas then the following are formulas: $(\alpha \wedge \beta)$, $(\alpha \vee \beta)$, $(\alpha \rightarrow \beta)$ *(etc.)*
- *Examples of formulas:*
  $(p1 \vee \neg p2)$, $\neg \neg \neg(p9 \rightarrow p8)$, $(p1 \wedge (p2 \vee p3))$
- *etc.*

The language of propositional logic defined more properly

- *Examples of formulas:*
  $(p1 \vee \neg p2)$, $\neg \neg \neg(p9 \rightarrow p8)$, $(p1 \wedge (p2 \vee p3))$
- *Convention 1:* outermost brackets are omitted,:
  $p1 \vee \neg p2$, $\neg \neg \neg(p9 \rightarrow p8)$, $p1 \wedge (p2 \vee p3)$
- *Convention 2:* associativity allows us to omit even more brackets, e.g.:
  $p1 \wedge p2 \wedge p3$, $p1 \vee p2 \vee p3$

Precedence of logical operators Nested Propositional Expressions

- Use parentheses to *group sub-expressions*: "I just saw my old *f*riend, and either he's *g*rown or I've *s*hrunk." = $f \wedge (g \vee s)$
  - $(f \wedge g) \vee s$ would mean something different
  - $f \wedge g \vee s$ would be ambiguous
- By convention, "$\neg$" takes *precedence* over both "$\wedge$" and "$\vee$".
  - $\neg s \wedge f$ means $(\neg s) \wedge f$, **not** $\neg (s \wedge f)$
- **1 $\neg$ 2 $\wedge$ $\vee$ 3 $\rightarrow$ $\leftrightarrow$**

Logic as shorthand for NL

Let p="It rained last night",
q="The sprinklers came on last night,"
r="The lawn was wet this morning."

"It didn't rain last night."

$\neg p$ =

"The lawn was wet this morning, and it didn't rain last night."

$r \wedge \neg p$ =

$\neg\ r \vee p \vee q$ =

"Either the lawn wasn't wet this morning, or it rained last night, or the sprinklers came on last night."

Truth Tables For Compound Propositions

- Construction of a truth table:
- Rows
  - Need a row for every possible combination of values for the atomic propositions.
- Columns
  - Need a column for the compound proposition (usually at far right)
  - Need a column for the truth value of each expression that occurs in the compound proposition as it is built up.
    - This includes the atomic propositions

Example Truth Table

- Construct a truth table for $p \vee q \rightarrow \neg r$

p	q	r	¬r	p ∨ q	p ∨ q → ¬r
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	T

Logic and bit operations

- Computer bit operations correspond to the logical connectives.
- Bitwise OR bitwise AND bitwise XOR

Bitwise Operations

- Boolean operations can be extended to operate on bit strings as well as single bits.
- E.g.:

```
01 1011 0110
11 0001 1101
-------------
11 1011 1111  Bit-wise OR
01 0001 0100  Bit-wise AND
10 1010 1011  Bit-wise XOR
```

§ 1.2 Applications of Propositional Logic

Applications of Propositional Logic: Summary

- Translating English to Propositional Logic
- System Specifications 系统规格
- Boolean Searching
- Logic Puzzles
- Logic Circuits

Translating sentences Example(命题翻译)

- ‘If I go to school or go home, I will not go shopping.’
  - P: I go to school
  - Q: I go home
  - R: I will go shopping
- If.....P.....or.....Q.....then....not.....R
  - $(P \vee Q) \rightarrow \neg R$
- You can access the Internet from campus only if you are a computer science major or you are not a freshman.
- $a \rightarrow (c \vee \neg f)$

Translating English Sentences

- Steps to convert an English sentence to a statement in propositional logic
  - Identify atomic propositions and represent using propositional variables.
  - Determine appropriate logical connectives
- "You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old."
  - *p*: you are under 4 feet tall,
  - q: you are older than 16 years old.
  - *r*: You can ride the roller coaster.

If *p* and ¬*q* then not *r*. $(p \wedge \neg q) \rightarrow \neg r$

Example

Problem: Translate the following sentence into propositional logic:

"You can access the Internet from campus only if you are a computer science major or you are not a freshman."

One Solution: Let a, c, and f represent respectively "You can access the internet from campus," "You are a computer science major," and "You are a freshman."

$$a \rightarrow (c \vee \neg f)$$

翻译:

她既聪明又用功

除非你努力,否则你将失败.

甲和乙中有一人且只有一人是大学生.

翻译

- 我和他既是弟兄又是同学。
- 别问我是谁.

System Specifications

- System and Software engineers take requirements in English and express them in a precise specification language based on logic.

  Example: Express in propositional logic:

  "The automated reply cannot be sent when the file system is full"

  Solution: One possible solution: Let *p* denote "The automated reply can be sent" and *q* denote "The file system is full."

$$q \rightarrow \neg p$$

Consistent System Specifications

Definition: A list of propositions is *consistent* if it is possible to assign truth values to the proposition variables so that each proposition is true.

Exercise: Are these specifications consistent?

- "The diagnostic message is stored in the buffer or it is retransmitted."
- "The diagnostic message is not stored in the buffer."
- "If the diagnostic message is stored in the buffer, then it is retransmitted."

Solution: Let p denote "The diagnostic message is not stored in the buffer." Let q denote "The diagnostic message is retransmitted" The specification can be written as: $p \vee q$, $p \rightarrow q$, $\neg p$. When p is false and q is true all three statements are true. So the specification is consistent.

- What if "The diagnostic message is not retransmitted is added."

  Solution: Now we are adding $\neg q$ and there is no satisfying assignment. So the specification is not consistent.

Logic Puzzles



Raymond Smullyan (Born 1919)

- An island has two kinds of inhabitants, *knights*, who always tell the truth, and *knaves*, who always lie.
- You go to the island and meet A and B.
  - A says "B is a knight."
  - B says "The two of us are of opposite types."

Example: What are the types of A and B?

Solution: Let p and q be the statements that A is a knight and B is a knight, respectively. So, then $\neg p$ represents the proposition that A is a knave and $\neg q$ that B is a knave.

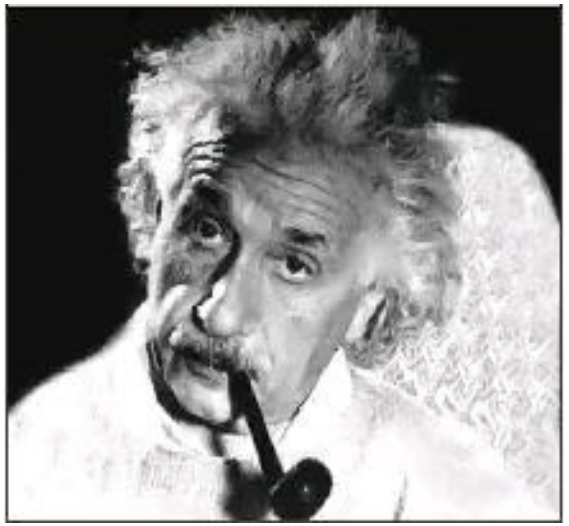
- If A is a knight, then p is true. Since knights tell the truth, q must also be true. Then $(p \wedge \neg q) \vee (\neg p \wedge q)$ would have to be true, but it is not. So, A is not a knight and therefore $\neg p$ must be true.
- If A is a knave, then B must not be a knight since knaves always lie. So, then both $\neg p$ and $\neg q$ hold since both are knaves.

Logic Puzzles

A father tells his two children, a boy and a girl, to play in their backyard without getting dirty.However, while playing, both children get mud on their foreheads. When the children stop playing, the father says “At least one of you has a muddy forehead,” and then asks the children to answer “Yes” or “No” to the question: “Do you know whether you have a muddy forehead?” The father asks this question twice. What will the children answer each time this question is asked, assuming that a child can see whether his or her sibling has a muddy forehead, but cannot see his or her own forehead? Assume that both children are honest and that the children answer each question simultaneously.

谁养鱼？

- 连爱因斯坦都大呼过瘾的题目!



谁养鱼？

- 此题源自 1981 年柏林的德国逻辑思考学院.....98%的测验者无法解题,以下是题目的中译版.
- 前题
  1. 有五间房屋排成一列
  2. 所有房屋的外表颜色都不一样
  3. 所有的屋主来自不同的国家
  4. 所有的屋主都养不同的宠物; 喝不同的饮料; 抽不同牌的香烟

谁养鱼

题示:

1. 英国人住在红色房屋
2. 瑞典人养狗
3. 丹麦人喝茶
4. 绿色的房子在白色房子的左边
5. 绿色房屋的屋主喝咖啡
6. 抽 Pall Mall 香烟的屋主养鸟
7. 黄色屋主抽 Dunhill
8. 位於最中间的屋主喝牛奶
9. 挪威人住在第一间房屋
10. 抽 Blend 的人住在养猫人家的隔壁
11. 养马的屋主隔壁住抽 Dunhill 的人家
12. 抽 Blue Master 的屋主他喝啤酒
13. 德国人他抽 Prince
14. 挪威人住在蓝色房子隔壁
15. 只喝开水的人家住在抽 Blend 的隔壁



谁养鱼？

- 问题: 请问谁养鱼？

- 房屋——红色、绿色、白色、黄色、蓝色
- 主人——英国、瑞典、丹麦、挪威、德国
- 饮料——茶、咖啡、牛奶、啤酒、水
- 宠物——狗、鸟、猫、马、鱼
- 香烟——Pall、Dunhill、Blend、Blue 、Prince

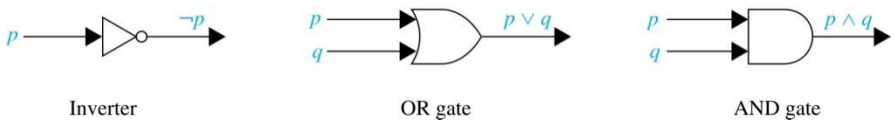
Question

- 上题中，缺少一个或多个条件，结果就不唯一。
- 如何得到多个（全部）答案？



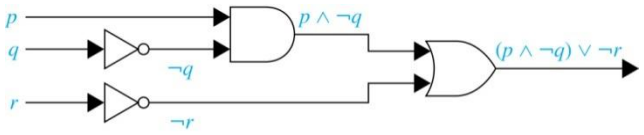
Logic Circuits

- Electronic circuits; each input/output signal can be viewed as a 0 or 1.
  - 0 represents **False**
  - 1 represents **True**
- Complicated circuits are constructed from three basic circuits called gates.



Inverter OR gate AND gate

  - The inverter (**NOT gate**)takes an input bit and produces the negation of that bit.
  - The **OR gate** takes two input bits and produces the value equivalent to the disjunction of the two bits.
  - The **AND gate** takes two input bits and produces the value equivalent to the conjunction of the two bits.
- More complicated digital circuits can be constructed by combining these basic circuits to produce the desired output given the input signals by building a circuit for each piece of the output expression and then combining them. For example:



作业

- § 1.1 2, 14,16,30,32,40
- § 1.2 4,22,36,44

1 The Foundations: Logic and Proofs

1.1 Propositional Logic

1.2Applications of Propositional Logic

1.3 Propositional Equivalences

1.4 Predicates and Quantifiers

1.5 Nested Quantifiers

1.6 Rules of Inference

1.7 Introduction to Proofs

1.8 Proof Methods and Strategy



Propositional Equivalences

Rosen 8th ed., §§1.3

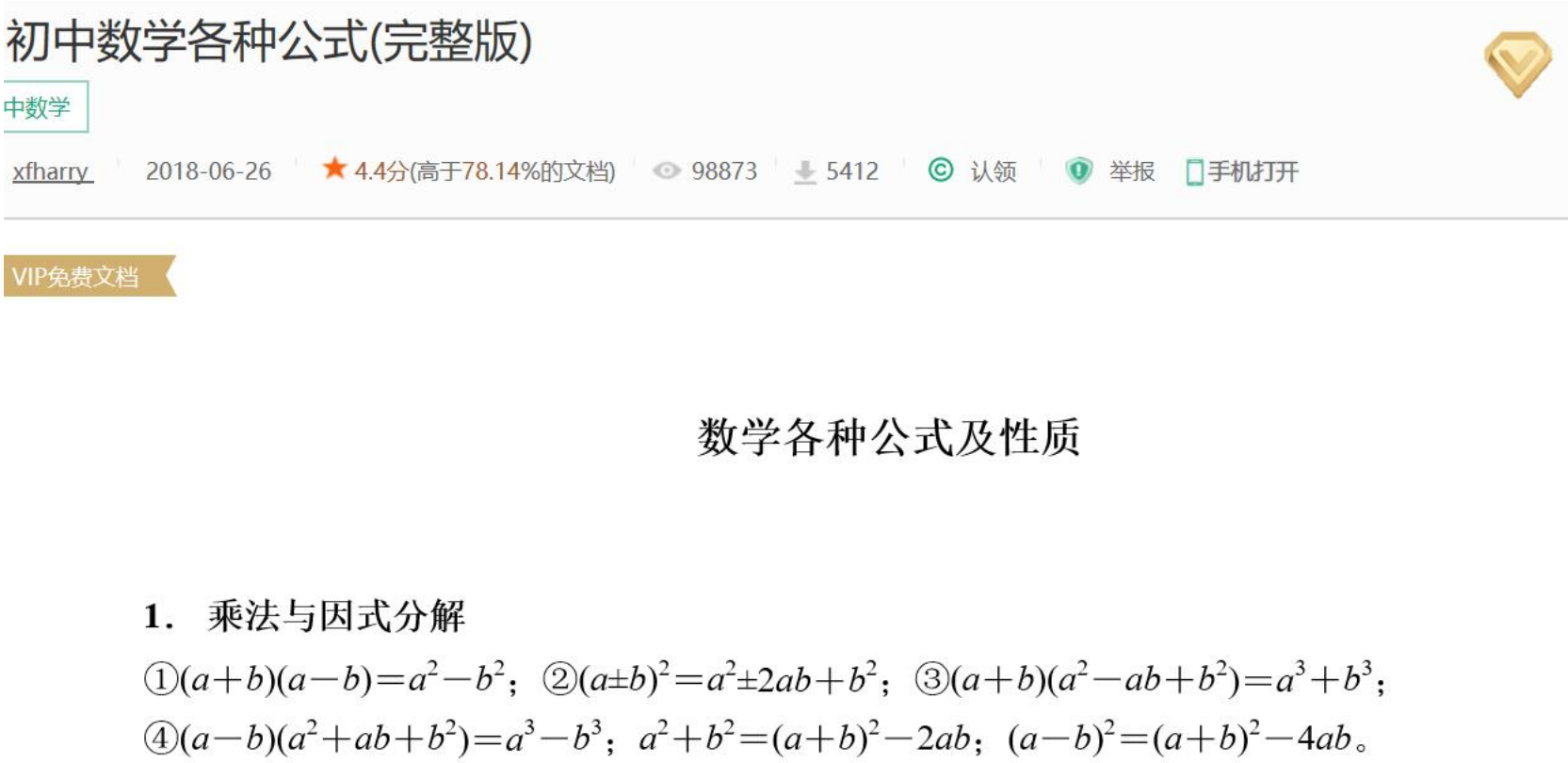
Some important ideas:

- Distinguishing between different kinds of formulas
- Seeing that some formulas that look different may express the same information

- First: different kinds of formulas

1.3 Propositional Equivalences

- 逻辑等价式
- Similar to arithmetic identities in algebra, but for propositional equivalences instead.



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数学各种公式及性质

1. 乘法与因式分解

①$(a+b)(a-b)=a^2-b^2$；②$(a\pm b)^2=a^2\pm 2ab+b^2$；③$(a+b)(a^2-ab+b^2)=a^3+b^3$；

④$(a-b)(a^2+ab+b^2)=a^3-b^3$；$a^2+b^2=(a+b)^2-2ab$；$(a-b)^2=(a+b)^2-4ab$。

Section Summary

- Tautologies, Contradictions, and Contingencies.
- Logical Equivalence
  - Important Logical Equivalences
  - Showing Logical Equivalence
- Normal Forms
  - Disjunctive Normal Form
  - Conjunctive Normal Form
- Propositional Satisfiability
  - The n-Queens Problem
  - Sudoku Example

Tautologies永真式

A *tautology* is a compound proposition that is **true** *no matter what* the truth values of its atomic propositions are!

Ex. $p \vee \neg p$ [What is its truth table?]

Tautologies

- When every row of the truth table gives T.

- Example:

p	$\vee$	$\neg$p
T	T	FT
F	T	TF

Contradictions永假式

A *contradiction* is a compound proposition that is **false** no matter what! *Ex.* $p \wedge \neg p$

[Truth table?]

Contradictions

- When every row of the truth table gives F

- Example: $p \wedge \neg p$

T	F	F	T
F	F	T	F

What's left besides Tautologies and Contradictions

All other props. are *contingencies*:可能式 可满足式

Some rows give T, others give F

Now: formulas that have the same meaning

Tautologies, Contradictions, and Contingencies

- A tautology is a proposition which is always true.
  - Example: $p \vee \neg p$
- A *contradiction* is a proposition which is always false.
  - Example: $p \wedge \neg p$
- A *contingency* is a proposition which is neither a tautology nor a contradiction, such as p

P	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Which of these are tautologies?

1. $p \rightarrow (q \rightarrow p)$
2. $p \rightarrow (\neg p \rightarrow p)$
3. $(q \rightarrow p) \rightarrow (p \rightarrow q)$
4. $(q \rightarrow p) \vee (p \rightarrow q)$
5. $(p \rightarrow (q \rightarrow r)) \rightarrow (q \rightarrow (p \rightarrow r))$

Please prove your claims, using truth tables. (Hint: Ask what assignment of truth values to p,q, and r would *falsify* each formula).

Which of these are tautologies?

1. $p \rightarrow (q \rightarrow p)$ *Tautologous*
2. $p \rightarrow (\neg p \rightarrow p)$ *Tautologous*
3. $(q \rightarrow p) \rightarrow (p \rightarrow q)$ *Contingent*
4. $(q \rightarrow p) \vee (p \rightarrow q)$ *Tautologous*
5. $(p \rightarrow (q \rightarrow r)) \rightarrow (q \rightarrow (p \rightarrow r))$ *Tautologous*

Logical Equivalence等价式

Compound proposition *p* is *logically equivalent* to compound proposition *q*, written $p \Leftrightarrow q$, **IFF**
p and *q* contain the same truth values in all rows of their truth tables

We will also say: they express the same truth function (= the same function from values for atoms to values for the whole formula).

Propositional Equivalence

Two *syntactically* (*i.e.,* textually) different compound propositions may be *semantically* identical (*i.e.,* have the same meaning). We call them *logically equivalent*. 逻辑等价

Notation: … ⇔ …

Proving Equivalence via Truth Tables

Ex. Prove that $p \vee q \Leftrightarrow \neg(\neg p \wedge \neg q)$.

p	q	$p \vee q$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$	$\neg(\neg p \wedge \neg q)$
F	F	F	T	T	T	F
F	T	T	T	F	F	T
T	F	T	F	T	F	T
T	T	T	F	F	F	T

Equivalence Laws

- Similar to arithmetic identities in algebra, but for propositional equivalences instead.
- They provide a pattern or template that can be used to match all or part of a much more complicated proposition and to find an equivalence for it.
- Abbreviation: **T** stands for an arbitrary tautology; **F** an arbitrary contradiction

Logical Equivalences(等价式)

- $P \wedge T \Leftrightarrow P$ Identity（同一律）
- $P \vee F \Leftrightarrow P$
- $P \vee T \Leftrightarrow T$ Domination（零律）
- $P \wedge F \Leftrightarrow F$
- $P \vee P \Leftrightarrow P$ Idempotency（等幂律）
- $P \wedge P \Leftrightarrow P$

Logical Equivalences(等价式)

- $\neg(\neg P) \Leftrightarrow P$ Double negation（双重否定律）
- $P \vee Q \Leftrightarrow Q \vee P$ Commutativity（交换律）
- $P \wedge Q \Leftrightarrow Q \wedge P$
- $(P \vee Q) \vee R \Leftrightarrow P \vee (Q \vee R)$ Associativity（结合律）
- $(P \wedge Q) \wedge R \Leftrightarrow P \wedge (Q \wedge R)$

Logical Equivalences

- $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$ Distributivity（分配律）
- $P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$
- $\neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$ DeMorgan's laws（德·摩根律）
- $\neg(P \vee Q) \Leftrightarrow \neg P \wedge \neg Q$

Logical Equivalences

- $P \to Q \Leftrightarrow \neg P \vee Q$ Implication（蕴涵等价式）
- $P \vee \neg P \Leftrightarrow T$ Tautology（排中律）
- $P \wedge \neg P \Leftrightarrow F$ Contradiction（矛盾律）
- $(P \to Q) \wedge (Q \to P) \Leftrightarrow (P \leftrightarrow Q)$ Equivalence
- $(P \to Q) \wedge (P \to \neg Q) \Leftrightarrow \neg P$ Absurdity 归谬论

Logical Equivalences

- $(P \rightarrow Q) \Leftrightarrow (\neg Q \rightarrow \neg P)$ Contrapositive（蕴涵逆反式）
- $P \vee (P \wedge Q) \Leftrightarrow P$, $P \wedge (P \vee Q) \Leftrightarrow P$ Absorption（吸收律）
- $(P \wedge Q) \rightarrow R \Leftrightarrow P \rightarrow (Q \rightarrow R)$
  Exportation（输出律）

What's the difference between $\leftrightarrow$ and $\Leftrightarrow$?

$A \Leftrightarrow B$ says that **no assignment of truth values** to A and B can make $A \leftrightarrow B$ false

So, $A \Leftrightarrow B$ can only hold between well-chosen compound A and B. For example,

$p \vee q \Leftrightarrow \neg(\neg p \wedge \neg q)$.

$A \leftrightarrow B$ says that A and B happen to have the same truth value. For example,

A=2+2=4. B=I teach at BUPT.

A=2+2=4. B=I teach at BUPT or BIT.

You have learned about:

- Propositional logic operators'
  - Symbolic notations.
  - English equivalents
  - Truth tables.
  - Logical equivalence

- Next:
  - More about logical equivalences.
  - How to prove them.

判断命题公式逻辑等价的方法：

- 1、真值表
- 2、命题公式的演算
- 基本等值定理；
- 代入规则
- 置换规则

代入规则

- 在一个永真公式中，任何一个命题变元R出现的每一处，用另一个命题公式代入，所得命题公式仍是永真式。

  - 例：求证： $(p \rightarrow q) \vee \neg (p \rightarrow q)$为永真式
  - $R \vee \neg R \Leftrightarrow T$
  - 用$(p \rightarrow q)$代入上式中所有的R

置换规则

- 给定公式A，它的所有子公式为A_1，A_2，…A_n ，设$A_i \Leftrightarrow B_i$, 用B_i置换A_i, 所得公式为B，则 $A \Leftrightarrow B$。
- 例：求证：
- $(P \rightarrow (Q \rightarrow R)) \Leftrightarrow P \rightarrow (\neg Q \vee R)$为永真式
- 已知$(Q \rightarrow R) \Leftrightarrow (\neg Q \vee R)$
- 用$(\neg Q \vee R)$置换原式中的$(Q \rightarrow R)$

命题公式逻辑等价关系的应用：

1、判定是否逻辑等价；

- 2、判断是否为永真公式或永假公式；
- 3、命题公式的化简

Equivalence Laws - Examples

- *Identity*: $p \wedge \mathbf{T} \Leftrightarrow p$ $p \vee \mathbf{F} \Leftrightarrow p$
- *Domination*: $p \vee \mathbf{T} \Leftrightarrow \mathbf{T}$ $p \wedge \mathbf{F} \Leftrightarrow \mathbf{F}$
- *Idempotence*: $p \vee p \Leftrightarrow p$ $p \wedge p \Leftrightarrow p$
- *Double negation:* $\neg\neg p \Leftrightarrow p$
- *Commutativity:* $p \vee q \Leftrightarrow q \vee p$ $p \wedge q \Leftrightarrow q \wedge p$
- *Associativity:* $(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r)$
  $(p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$

More Equivalence Laws

- *Distributive*: $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$
  $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$
- *De Morgan's*:
  $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$
  $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$
- *Trivial tautology/contradiction*:
  $p \vee \neg p \Leftrightarrow \mathbf{T}$ $p \wedge \neg p \Leftrightarrow \mathbf{F}$



Augustus
De Morgan
(1806-1871)

Defining Operators via Equivalences

Using equivalences, we can *define* operators in terms of other operators.

- Exclusive or: $p \oplus q \Leftrightarrow (p \vee q) \wedge \neg(p \wedge q)$
  $p \oplus q \Leftrightarrow (p \wedge \neg q) \vee (q \wedge \neg p)$
- Implies: $p \rightarrow q \Leftrightarrow \neg p \vee q$
- Biconditional: $p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$
  $p \leftrightarrow q \Leftrightarrow \neg(p \oplus q)$

Tautologies revisited

- We've introduced the notion of a tautology using the example $p \vee \neg p$
- Now, you know more operators, so can formulate many more tautologies, e.g.,
  $(p \vee q) \leftrightarrow \neg(\neg p \wedge \neg q)$
  $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$, *and so on*

EXAMPLE 4

- Show that the propositions $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent.This is the distributive law of disjunction over

Solution:

We construct the truth table for these propositions in Table 4. Since the truth values of $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ agree, these propositions are logically equivalent.

Table 4

TABLE 4

A Demonstration That $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ Are Logically Equivalent.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Constructing New Logical Equivalences

- We can show that two expressions are logically equivalent by developing a series of logically equivalent statements.
- To prove that $A \equiv B$ we produce a series of equivalences beginning with A and ending with B.

$$A \equiv A_1$$
$$\vdots$$
$$A_n \equiv B$$

- Keep in mind that whenever a proposition (represented by a propositional variable) occurs in the equivalences listed earlier, it may be replaced by an arbitrarily complex compound proposition.

Equivalence Proofs

Example: Show that $\neg(p \vee (\neg p \wedge q))$ is logically equivalent to $\neg p \wedge \neg q$

Solution:

$\neg(p \vee (\neg p \wedge q))$	$\equiv$	$\neg p \wedge \neg(\neg p \wedge q)$	by the second De Morgan law
	$\equiv$	$\neg p \wedge [\neg(\neg p) \vee \neg q]$	by the first De Morgan law
	$\equiv$	$\neg p \wedge (p \vee \neg q)$	by the double negation law
	$\equiv$	$(\neg p \wedge p) \vee (\neg p \wedge \neg q)$	by the second distributive law
	$\equiv$	$F \vee (\neg p \wedge \neg q)$	because $\neg p \wedge p \equiv F$
	$\equiv$	$(\neg p \wedge \neg q) \vee F$	by the commutative law for disjunction
	$\equiv$	$(\neg p \wedge \neg q)$	by the identity law for **F**

Equivalence Proofs

Example: Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Solution:

$(p \wedge q) \rightarrow (p \vee q)$	$\equiv$	$\neg(p \wedge q) \vee (p \vee q)$	by truth table for $\rightarrow$
	$\equiv$	$(\neg p \vee \neg q) \vee (p \vee q)$	by the first De Morgan law
	$\equiv$	$(\neg p \vee p) \vee (\neg p \vee \neg q)$	by associative and commutative laws
			laws for disjunction
	$\equiv$	$T \vee T$	by truth tables
	$\equiv$	T	by the domination law

An Example Problem

- Check using a symbolic derivation whether $(p \wedge \neg q) \rightarrow (p \oplus r) \Leftrightarrow \neg p \vee q \vee \neg r$.

$(p \wedge \neg q) \rightarrow (p \oplus r)$ [Expand definition of $\rightarrow$]

$\Leftrightarrow \neg(p \wedge \neg q) \vee (p \oplus r)$ [Expand defn. of $\oplus$]

$\Leftrightarrow \neg(p \wedge \neg q) \vee ((p \vee r) \wedge \neg(p \wedge r))$

[DeMorgan's Law]

$\Leftrightarrow (\neg p \vee q) \vee ((p \vee r) \wedge \neg(p \wedge r))$

cont.

Example Continued...

$(\neg p \vee q) \vee ((p \vee r) \wedge \neg(p \wedge r)) \Leftrightarrow$ [$\vee$ commutes]

$\Leftrightarrow (q \vee \neg p) \vee ((p \vee r) \wedge \neg(p \wedge r))$ [$\vee$ is associative]

$\Leftrightarrow q \vee (\neg p \vee ((p \vee r) \wedge \neg(p \wedge r)))$ [distrib. $\vee$ over $\wedge$]

$\Leftrightarrow q \vee (((\neg p \vee (p \vee r)) \wedge (\neg p \vee \neg(p \wedge r)))$

[assoc.] $\Leftrightarrow q \vee (((\neg p \vee p) \vee r) \wedge (\neg p \vee \neg(p \wedge r)))$

[trivial taut.] $\Leftrightarrow q \vee ((\mathbf{T} \vee r) \wedge (\neg p \vee \neg(p \wedge r)))$

[domination] $\Leftrightarrow q \vee (\mathbf{T} \wedge (\neg p \vee \neg(p \wedge r)))$

[identity] $\Leftrightarrow q \vee (\neg p \vee \neg(p \wedge r)) \Leftrightarrow$ *cont.*

End of Long Example

$q \vee (\neg p \vee \neg(p \wedge r))$

[DeMorgan's] $\Leftrightarrow q \vee (\neg p \vee (\neg p \vee \neg r))$

[Assoc.] $\Leftrightarrow q \vee ((\neg p \vee \neg p) \vee \neg r)$

[Idempotent] $\Leftrightarrow q \vee (\neg p \vee \neg r)$

[Assoc.] $\Leftrightarrow (q \vee \neg p) \vee \neg r$

[Commut.] $\Leftrightarrow \neg p \vee q \vee \neg r$

Q.E.D. (quod erat demonstrandum)

Q.E.D.是拉丁片语「quod erat demonstrandum」（这被证明了）的缩写。

(Which was to be shown.)

练习

- 求证： $Q \vee \neg((\neg P \vee Q) \wedge P)$是永真式。

Logically Equivalent

- Two compound propositions p and q are logically equivalent if $p \leftrightarrow q$ is a tautology.
- We write this as $p \Leftrightarrow q$ or as $p \equiv q$ where p and q are compound propositions.
- Two compound propositions p and q are equivalent if and only if the columns in a truth table giving their truth values agree.
- This truth table show $\neg p \vee q$ is equivalent to $p \rightarrow q$.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

De Morgan's Laws



$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Augustus De Morgan

1806-1871

This truth table shows that De Morgan's Second Law holds.

p	q	$\neg p$	$\neg q$	$(p \vee q)$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

Key Logical Equivalences

- Identity Laws:

$$p \wedge T \equiv p \qquad p \vee F \equiv p$$

- Domination Laws:

$$p \vee T \equiv T \qquad p \wedge F \equiv F$$

- Idempotent laws:

$$p \vee p \equiv p \qquad p \wedge p \equiv p$$

- Double Negation Law:

$$\neg(\neg p) \equiv p$$

- Negation Laws:

$$p \vee \neg p \equiv T \qquad p \wedge \neg p \equiv F$$

Key Logical Equivalences (*cont*)

- Commutative Laws: ,

$$p \vee q \equiv q \vee p \quad p \wedge q \equiv q \wedge p$$

- Associative Laws:

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$
$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

- Distributive Laws:

$$(p \vee (q \wedge r) \equiv (p \vee q)) \wedge (p \vee r)$$
$$(p \wedge (q \vee r)) \equiv (p \wedge q) \vee (p \wedge r)$$

- Absorption Laws:

$$p \vee (p \wedge q) \equiv p \quad p \wedge (p \vee q) \equiv p$$

More Logical Equivalences

TABLE 7 **Logical Equivalences Involving Conditional Statements.**

$$p \to q \equiv \neg p \vee q$$

$$p \to q \equiv \neg q \to \neg p$$

$$p \vee q \equiv \neg p \to q$$

$$p \wedge q \equiv \neg(p \to \neg q)$$

$$\neg(p \to q) \equiv p \wedge \neg q$$

$$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$$

$$(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$$

$$(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$$

$$(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$$

TABLE 8 **Logical Equivalences Involving Biconditional Statements.**

$$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

简单析取式和简单合取式

- 仅有有限个命题变元或其否定的析取构成的析取式称为简单析取式，或基本和。
  - 例如：$\neg P \vee Q \vee R$，$\neg P \vee \neg Q \vee P$是简单析取式
- 而仅有有限个命题变元或其否定的合取构成的合取式称为简单合取式，或基本积。
  - 例如：$\neg P \wedge R$，$Q \wedge R \wedge \neg Q$是简单合取式。
- P，$\neg Q$一个变项本身可以看作简单析取式也可以看作简单合取式。
- 而$\neg P \vee （Q \wedge R）$，$（P \wedge Q）\vee \neg R$不是简单析取式。

析取范式

- 析取范式：由有限个简单合取式（基本积）的析取构成的析取式称为析取范式。
- 例
  - $P\vee(P\wedge Q)\vee(\neg P\wedge\neg Q\wedge\neg R)$是析取范式
    - $P\vee Q\vee R$是析取范式，是三个简单合取范式的析取

合取范式

- 由有限个简单析取式（基本和）的合取构成的合取式称为合取范式。
- 例：
  - （P∨Q）∧¬Q∧（Q∨¬R∨S）是合取范式
  - P∨Q∨R是合取范式，是一个简单析取范式的合取

范式的存在性定理

- 任意一个命题公式均存在一个与之等值的析取范式和合取范式
- 证明：对于任意公式，可用下面的方法构造出与其等价的析取范式和合取范式
  - 第一步： 利用等价式，消去公式中的联结符→和↔
    - ⑴ $P \to Q \Leftrightarrow \neg P \vee Q$
    - ⑵ $P \leftrightarrow Q \Leftrightarrow （P \to Q）\wedge（Q \to P）$
  - 第二步：利用德摩根律将公式中的联结符¬移到命题变元之前，用双重否定律消去两个连续的联结符¬
  - 第三步：用分配律将公式化为基本积的析取或基本和的合取。

例

- 例2：求公式（P ∨ Q） → R的析取范式和合取范式
- 解：
  - （P ∨ Q） → R ⇔ ¬（P ∨ Q） ∨ R ⇔ （¬P ∧ ¬Q） ∨ R⇔（¬P ∨ R） ∧ （¬Q ∨ R）合取范式
  - 原式⇔（¬P ∨ R） ∧ （¬Q ∨ R） ⇔（ （¬P ∨ R） ∧ （¬Q ）） ∨ （ （¬P ∨ R） ∧ R） ⇔ （¬P ∧ ¬ Q） ∨ （R ∧ ¬Q） ∨ （¬P ∧ R） ∨ R 析取范式
  - 原式⇔ （¬P ∧ ¬Q） ∨ （R ∧ ¬Q） ∨ R ⇔ （¬P ∧ ¬Q） ∨ R 析取范式
  - 析取（合取）范式不唯一！

$P \vee (P \wedge Q) \Leftrightarrow P$, $P \wedge (P \vee Q) \Leftrightarrow P$
Absorption（吸收律）

- 令A（a1、a2、……、an）包含有n个变量的公式，
- 极小项(extremal ~):小项中恰包含n个变量或其否定。
- 极大项(extremal ~):大项中恰包含n个变量或其否定。
- **主合取范式**（Major conjunctive normal form):
- **principal** conjunctive normal form
- 若干个极大项的合取。
- **主析取范式**（Major disjunctive normal form):

principal disjunctive normal form

- 若干个极小项的析取。

极小项

- 如公式中共有**N**个命题变项**P**，……，P_n这**N**个变项的合取式中，每个变项P_i或其否定$\neg P_i$，均出现且两者仅出现一个，并且按命题变项的下标排列（字母按字典序列）这样的简单合取式称为极小项，又称布尔合取。

极小项

- 例如：公式中出现P，Q，R三个命题变项

  P∧¬Q∧R，¬P∧Q∧¬R是极小项

  P∧¬Q不是极小项，其中没出现R

  P∧¬Q∧¬ P不是极小项，P，¬P同时出现

极小项和赋值的一一对应

- 每一个极小项只有一个赋值使其为真，其余2^n-1个赋值使其为假。
- 例如：
  - 公式中出现P，Q，R三个不同的赋值。极小项$\neg P \wedge Q \wedge \neg R$，赋值010使其为真，其余$2^3$-1=7个赋值使其为假。这样极小项与赋值建立了一一对应关系。
  - 而$\neg P \wedge Q$不是极小项，有两个赋值010，011使其为真，其余6个赋值使其为假。

极小项的记法

- 公式共有2^n个赋值，每个对应赋值一个极小项。如极小项对应赋值$a_1a_2a_3$，把其看作二进数，化为十进数为$k_0 \le k \le 2^n-1$,则该极小项记作m_{a1a2a3},也记作m_k

极小项	对应的赋值	十进制数	记法
¬P∧¬Q∧¬R	000	0	m_{000}或m_0
¬P∧¬Q∧R	001	1	m_{001}或m_1
¬P∧Q∧¬R	010	2	m_{010}或m_2
¬P∧Q∧R	011	3	m_{011}或m_3
P∧¬Q∧¬R	100	4	m_{100}或m_4
P∧¬Q∧R	101	5	m_{101}或m_5
P∧Q∧¬R	110	6	m_{110}或m_6
P∧Q∧R	111	7	m_{111}或m_7

主析取范式

Principal disjunctive normal form

principal disjunctive normal form

- 若干个不同的极小项的析取式，称为主析取范式

定理

- 任何一个命题公式均存在一个与之等值的主析取范式，而且是唯一的。
- 例如：
  - $P\vee(\neg P\wedge Q)$，$(P\wedge Q)\vee(\neg P\wedge R)$是析取范式,但不是主析取范式。
    - $(P\wedge Q\wedge\neg R)\vee(\neg P\wedge Q\wedge R)$是主析取范式，当然也是析取范式。

求主析取范式方法之一：真值表法

- 一个公式的主析取范式与该公式的真值表是一一对应的
- 例如：由三个极小项组成的主析取范式

（P∧Q∧R）∨（¬P∧Q∧R）∨（¬P∧Q∧¬R）

  - 这三个极小项对应的赋值分别是111，011，010，则该三个赋值使公式为真，而其它五个赋值使公式为假。
    - 反过来，如果这三个赋值使公式为真，而其它赋值使公式为假，则该公式的主析取范式就是由这三个赋值对应的极小项析取而成的。

例：用真值表的方法求¬（P∧Q）的主析取范式

P Q	P∧Q	¬（P∧Q）
0 0	0	1
0 1	0	1
1 0	0	1
1 1	1	0

- 解：
  - ¬（P∧Q）的真值表
  - 找使公式为真的赋值。
  - 00对应的极小项是¬P∧¬Q （m_0）
  - 01对应的极小项是¬P∧Q （m_1）
  - 10对应的极小项是P∧¬Q （m_2）
  - 则原公式的主析取范式是
  - ¬（P∧Q）⇔（¬P∧¬Q）∨（¬P∧Q）∨（P∧¬Q）
    ⇔ $m_0 \vee m_1 \vee m_2$

求主析取范式的方法之二：等值演算法

- 利用十六个命题公式、命题定理用等值演算的方法一步步将它化为主析取范式。
- 第一步：将公式化为析取范式，“消去”含有矛盾式的简单合取式，这是因为（$P\wedge\neg P\wedge Q$）$\vee G\Leftrightarrow G$，而$P\wedge P$用P代替。
- 第二步：若析取范式中的简单合取式项不是极小项，即该项中缺命题变项，则添加（$P_i\vee\neg P_i$）再用分配律展开。
  - 例如：公式中含P、Q、R，某合取项为$P\wedge\neg R$，缺Q。则$P\wedge\neg R\Leftrightarrow P\wedge\neg R\wedge(Q\vee\neg Q)\Leftrightarrow(P\wedge Q\wedge\neg R)\vee(P\wedge\neg Q\wedge\neg R)$
- 第三步：用幂等律将重复的极小项删去，即$m_i\vee m_i\Leftrightarrow m_i$然后将各极小项按顺序排列。

例：求（（P∨Q）→R）→P的主析取范式

- 解：原公式⇔¬（¬（P∨Q）∨R）∨P
- ⇔（（P∨Q）∧¬R）∨P
- ⇔（P∧¬R）∨（Q∧¬R）∨P（析取范式）
- ⇔P∨（Q∧¬R）（析取范式）
- ⇔（P∧（Q∨¬Q）∧（R∨¬R））∨（（Q∧¬R）∧（P∨¬P））
- ⇔（P∧Q∧R）∨（P∧Q∧¬R）∨（P∧¬Q ∧R）∨（P∧¬Q ∧¬R）∨（P∧Q∧¬R）∨（¬P∧Q∧¬R）
- （六个极小项，其中重复了一个）
- ⇔（P∧Q∧R）∨（P∧Q∧¬R）∨（P∧¬Q ∧R）∨（P∧¬Q ∧¬R）∨（¬P∧Q∧¬R）
- ⇔m_2∨ m_4∨ m_5∨ m_6∨ m_7 ⇔Σ(2,4,5,6,7)

Normal or *Canonical* Forms(范式）

- Unique representations of a proposition
- Examples:
- Construct a simple proposition of two variables which is true only when
  - P is true and Q is false:

    $P \wedge \neg Q$
  - P is true and Q is true:

    $P \wedge Q$
  - P is true and Q is false or P is true and Q is true:

    $(P \wedge \neg Q) \vee (P \wedge Q)$

Normal or *Canonical* Forms

- A disjunction of conjunctions where
  - every variable or its negation is represented once in each conjunction （合取式）
    (a *minterm*)
  - each minterms appears only once

Disjunctive Normal Form (DNF)（析取范式）

- Important in switching theory, simplification in the design of circuits.
- Method: To find the minterms of the DNF.
  - Use the rows of the truth table where the proposition is 1 or True
  - If a zero appears under a variable, use the negation of the propositional variable in the minterm
  - If a one appears, use the propositional variable.

Example

- Find the DNF of $(P \vee Q) \rightarrow \neg R$

P	Q	R	$(P \vee Q) \rightarrow \neg R$
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Example

- There are 5 cases where the proposition is true, hence 5 minterms. Rows 1,2,3, 5 and 7 produce the following disjunction of minterms:

  $(P \vee Q) \rightarrow \neg R$

$\Leftrightarrow(\neg P \wedge \neg Q \wedge \neg R) \vee (\neg P \wedge \neg Q \wedge R) \vee (\neg P \wedge Q \wedge \neg R)$
$\vee (P \wedge \neg Q \wedge \neg R) \vee (P \wedge Q \wedge \neg R)$

  - Note that you get a *Conjunctive Normal Form* (CNF)（合取范式） if you negate a DNF and use DeMorgan's Laws.

主合取范式
（principal conjunctive normal form）:

- 若干个极大项的合取。
- **如公式中共有N个命题变项P，……，P_n这N个变项的析取式中，每个变项P_i或其否定$\neg P_i$，均出现且两者仅出现一个，并且按命题变项的下标排列（字母按字典序列）这样的简单析取式称为极大项。**

极大项的记法

极大项	对应的赋值	十进制数	记法
$P \vee Q \vee R$	000	0	M_{000}或M_0
$P \vee Q \vee \neg R$	001	1	M_{001}或M_1
$P \vee \neg Q \vee R$	010	2	M_{010}或M_2
$P \vee \neg Q \vee \neg R$	011	3	M_{011}或M_3
$\neg P \vee Q \vee R$	100	4	M_{100}或M_4
$\neg P \vee Q \vee \neg R$	101	5	M_{101}或M_5
$\neg P \vee \neg Q \vee R$	110	6	M_{110}或M_6
$\neg P \vee \neg Q \vee \neg R$	111	7	M_{111}或M_7

例：用真值表的方法求¬（P∧Q）的主合取范式

- 解：
  - ¬（P∧Q）的真值表
  - 找使公式为假的赋值。
  - 11对应的极大项是
  - $\neg P \vee \neg Q$ （M_3）

  - 则原公式的主合取范式是
  - $\neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q \Leftrightarrow M_3$

P Q	P∧Q	¬（P∧Q）
0 0	0	1
0 1	0	1
1 0	0	1
1 1	1	0

求主合取范式的方法之二：等值演算法

- 利用十六个命题公式、命题定理用等值演算的方法一步步将它化为主合取范式。
- 第一步：将公式化为合取范式，“消去”含有永真式的简单析取式，这是因为（$P \vee \neg P \vee Q$）$\wedge G \Leftrightarrow G$，而$P \vee P$用P代替。
- 第二步：若合取范式中的简单析取式项不是极大项，即该项中缺命题变项，则添加（$P_i \wedge \neg P_i$）再用分配律展开。
  - 例如：公式中含P、Q、R，某析取项为$P \vee \neg R$，缺Q。则$P \vee \neg R \Leftrightarrow P \vee \neg R \vee (Q \wedge \neg Q) \Leftrightarrow (P \vee Q \vee \neg R) \wedge (P \vee \neg Q \vee \neg R)$
- 第三步：用幂等律将重复的极大项删去，即$M_i \vee M_i \Leftrightarrow M_i$然后将各极大项按顺序排列。

Conjunctive Normal Form

Example: Put the following into CNF:

$$\neg(p \to q) \vee (r \to p)$$

Solution:

1. Eliminate implication signs:

$\Leftrightarrow \quad \neg(\neg p \vee q) \vee (\neg r \vee p)$

2. Move negation inwards; eliminate double negation:

$\Leftrightarrow \quad (p \wedge \neg q) \vee (\neg r \vee p)$

3. Convert to CNF using associative/distributive laws

$\Leftrightarrow \quad (p \vee \neg r \vee q) \wedge (\neg q \vee \neg r \vee p)$

$\Leftrightarrow (p \vee q \vee \neg r) \wedge (p \vee \neg q \vee \neg r) \Leftrightarrow M_1 \wedge M_3 \Leftrightarrow \prod(1, 3)$

Propositional Satisfiability

- A compound proposition is *satisfiable* if there is an assignment of truth values to its variables that make it true. When no such assignments exist, the compound proposition is *unsatisfiable*.
- A compound proposition is unsatisfiable if and only if its negation is a tautology.

Questions on Propositional Satisfiability

Example: Determine the satisfiability of the following compound propositions:

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$$

Solution: Satisfiable. Assign **T** to p, q, and r.

$$(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Solution: Satisfiable. Assign **T** to p and $\boldsymbol{F}$ to q.

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Solution: Not satisfiable. Check each possible assignment of truth values to the propositional variables and none will make the proposition true.

Notation

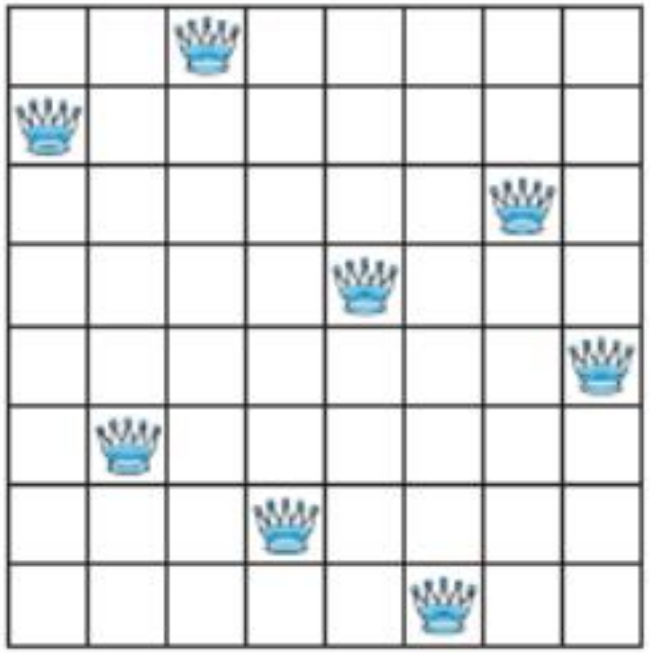
$\bigvee_{j=1}^{n} p_j$ is used for $p_1 \vee p_2 \vee \ldots \vee p_n$

$\bigwedge_{j=1}^{n} p_j$ is used for $p_1 \wedge p_2 \wedge \ldots \wedge p_n$

Needed for the next example.

EXAMPLE 10

- The n-Queens Problem The n-queens problem asks for a placement of n queens on an n × n chessboard so that no queen can attack another queen. This means that no two queens can be placed in the same row, in the same column, or on the same diagonal



- Every row contains at least one queen:

$$Q_1 = \bigwedge_{i=1}^{n} \bigvee_{j=1}^{n} p(i,j)$$

- There is at most one queen in each row

$$Q_2 = \bigwedge_{i=1}^{n} \bigwedge_{j=1}^{n-1} \bigwedge_{k=j+1}^{n} (\neg p(i,j) \vee \neg p$$ (i,k))

- No column contains more than one queen

$$Q_3 = \bigwedge_{j=1}^{n} \bigwedge_{i=1}^{n-1} \bigwedge_{k=i+1}^{n} (\neg p(i,j) \vee \neg p(k,j)).$$

- No diagonal contains two queens

$$Q_4 = \bigwedge_{i=2}^{n} \bigwedge_{j=1}^{n-1} \bigwedge_{k=1}^{\min(i-1,n-j)} (\neg p(i,j) \vee \neg p(i-k,k+j))$$

$$Q_5 = \bigwedge_{i=1}^{n-1} \bigwedge_{j=1}^{n-1} \bigwedge_{k=1}^{\min(n-i,n-j)} (\neg p(i,j) \vee \neg p(i+k,j+k))$$

$$Q = Q_1 \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5$$

Using this and other approaches, the number of ways n queens can be placed on a chessboard so that no queen can attack another has been computed for n ≤ 27. When n = 8 there are 92 such placements, while for n = 16 this number grows to 14,772,512.

P(1,1)	P(1,2)	P(1,3)	P(1,4)
P(2,1)	P(2,2)	P(2,3)	P(2,4)
P(3,1)	P(3,2)	P(3,3)	P(3,4)
P(4,1)	P(4,2)	P(4,3)	P(4,4)

$$Q_4 = \bigwedge_{i=2}^{n} \bigwedge_{j=1}^{n-1} \bigwedge_{k=1}^{\min(i-1,n-j)} (\neg p(i,j) \vee \neg p(i-k, k+j))$$

$$Q_5 = \bigwedge_{i=1}^{n-1} \bigwedge_{j=1}^{n-1} \bigwedge_{k=1}^{\min(n-i,n-j)} (\neg p(i,j) \vee \neg p(i+k, j+k))$$

Sudoku

- A **Sudoku puzzle** is represented by a 9×9 grid made up of nine 3×3 subgrids, known as **blocks**. Some of the 81 cells of the puzzle are assigned one of the numbers 1,2, …, 9.
- The puzzle is solved by assigning numbers to each blank cell so that every row, column and block contains each of the nine possible numbers.
- Example

	2	9				4		
			5			1		
	4							
				4	2			
6							7	
5								
7			3					5
	1			9				
							6	

Encoding as a Satisfiability Problem

- Let $p(i,j,n)$ denote the proposition that is true when the number n is in the cell in the ith row and the jth column.
- There are $9 \times 9 \times 9 = 729$ such propositions.
- In the sample puzzle $p(5,1,6)$ is true, but $p(5,j,6)$ is false for $j = 2,3,\ldots 9$

Encoding (cont)

- For each cell with a given value, assert $p(i,j,n)$, when the cell in row i and column j has the given value.
- Assert that every row contains every number.

$$\bigwedge_{i=1}^{9} \bigwedge_{n=1}^{9} \bigvee_{j=1}^{9} p(i,j,n)$$

$$\bigwedge_{j=1}^{9} \bigwedge_{n=1}^{9} \bigvee_{i=1}^{9} p(i,j,n)$$

- Assert that every column contains every number.

Encoding (cont)

- Assert that each of the 3 x 3 blocks contain every number.

$$\bigwedge_{r=0}^{2} \bigwedge_{s=0}^{2} \bigwedge_{n=1}^{9} \bigwedge_{i=1}^{3} \bigvee_{j=1}^{3} p(3r+i, 3s+j, n)$$

(this is tricky - ideas from chapter 4 help)

- Assert that no cell contains more than one number. Take the conjunction over all values of *n*, *n'*, *i*, and j, where each variable ranges from 1 to 9 and $n \neq n'$, of $p(i, j, n) \rightarrow \neg p(i, j, n')$

Solving Satisfiability Problems

- To solve a Sudoku puzzle, we need to find an assignment of truth values to the 729 variables of the form *p(i,j,n)* that makes the conjunction of the assertions true. Those variables that are assigned T yield a solution to the puzzle.
- A truth table can always be used to determine the satisfiability of a compound proposition. But this is too complex even for modern computers for large problems.
- There has been much work on developing efficient methods for solving satisfiability problems as many practical problems can be translated into satisfiability problems.

习题

- §1.3 – 8, 28, 38, 40, 44,52, 58,62
- 求下列公式的析取范式和合取范式
  - $((\neg A \vee \neg B) \rightarrow (A \leftrightarrow C)) \rightarrow B$
- 通过求主合取范式的方法，判断下列各对公式是否等价
  - $\neg A \vee (A \wedge B) \vee C$
  - $(\neg A \vee B) \wedge (B \vee C)$
- 通过求主析取范式，求出所有使下列公式为真的赋值
  - $(A \vee B) \wedge (A \rightarrow C) \wedge (B \rightarrow C)$

习题

- P，Q，R，S四个字母，从中取两个字母，但要满足下述三个条件，问有几种取法，如何取？
  - 如果取P，则R和S要取一个
  - Q，R不能同时取
  - 取R则不能取S

可选习题

A∧¬B∧C∧D∧¬E

A∧B∧¬C∧¬D∧E

- 张先生手中有代号为A、B、C、D、E的五种股票，根据当前股市情况及张先生本人的经济需求，需要有若干个股票抛出，但又必须满足如下复杂的要求：
- （1）若A抛出，则B也抛出；
- （2）B和C要留一种股票且只能留一种；
- （3）C和D要么全抛，要么都不抛；
- （4）D和E两种股票中必然有一种或两种要抛出；
- （5）若E抛出，则A、B也抛出。上述五种条件全部满足，问有几种合理的方案供张先生选择。

1 The Foundations: Logic and Proofs

- **1.1** Propositional Logic
- 1.2Applications of Propositional Logic
  1.3 Propositional Equivalences
  1.4 Predicates and Quantifiers
  1.5 Nested Quantifiers
  1.6 Rules of Inference
  1.7 Introduction to Proofs
  1.8 Proof Methods and Strategy

Foundations of Logic

Rosen 8th ed., §§1.4-1.5

Section Summary

- Predicates
- Variables
- Quantifiers
  - Universal Quantifier
  - Existential Quantifier
- Negating Quantifiers
  - De Morgan's Laws for Quantifiers
- Translating English to Logic
- Logic Programming (*optional*)

Propositional Logic Not Enough

- If we have:

  “All men are mortal.”

  “Socrates is a man.”
- Does it follow that “Socrates is mortal?”
- Can’t be represented in propositional logic. Need a language that talks about objects, their properties, and their relations.
- Later we’ll see how to draw inferences.
- “全部的男人是凡人. 苏格拉底是男人. 因此, 苏格拉底是凡人.”

Predicate Logic (§1.4)
谓词逻辑

- 前两节介绍的命题与命题演算是命题逻辑的内容，其基本组成单位是原子命题。一般地，原子命题作为具有真假意义的句子至少由主语和谓语两部分组成。
- 例如，计算机学院是北京邮电大学的一个下属学院。这里“计算机学院”是主语，而“是……”是谓语。当主语改变为“国际学院”时就得到新的原子命题：国际学院是北京邮电大学的一个下属学院。

- 由此可知，主语是独立存在的个体/值/Value，而谓语用来描述该个体的性质或个体间的关系，这里我们称其为谓词/Predicate。用P表示谓词“是……”。则P（计算机学院）或P（国际学院）分别等值于前述两个命题的表达。将个体用变量（称为个体变量）x推广，则P（x）表示：x是北京邮电大学的一个下属学院。这时该语句就不是一个命题，而是一个命题函数。

DEFINITION 1.

- 定义　一个谓词P连同相关的n（$n \geq 0$）个个体变量组成的表达式称为n元谓词（n-predicate），记$P(x_1, x_2, \ldots, x_n)$，其中n是该表达式中不同个体变量的数目。n元谓词也称简单命题函数，
- 复合命题函数　简单命题函数和复合命题函数，统称为命题函数（proposition function）。

EXAMPLE 1

- Let P(x) denote the statement "x > 3." What are the truth values of P(4) and P(2)?

```
P (4) = .T.

P (2) = .F.
```

EXAMPLE 2

- Let Q(x, y) denote the statement "x = y + 3." What are the truth values of the propositions Q(1, 2) and Q(3, 0)?

```
Q(1,2)= .F.

Q(3,0)= .T.
```

- 当$n>1$时，通常P给出了x_i $(i=1, 2, \ldots, n)$之间的关系。
- 例如，$P(x, y, z)$表示x位于y与z之间，是一个三元谓词。与x, y, z分别用教工食堂、运动场、图书馆代入时得到命题：教工食堂位于运动场与图书馆之间，其命值真值为T。（沙河校区）再如将杭州、南京、北京代入，则得到：杭州位于南京和北京之间，真值为F。
- 与$n=0$时（即0元谓词），命题函数就对应一个命题。

北京邮电大学沙河校区示意图



北邮人团队出品
2022.08

- 为了进一步讨论命题函数P(x)的真值情况，首先需要指定个体变量x可选择的范围，即个体域（universe of discourse, or domain）。每一个个体变量x都有自己的个体域。如果没有特别指定的个体域，则缺省为一个全个体域（total universe of discourse）即任意个体均可以作为常量对x代入。

- 在指定个体变量x的个体域后，该个体域中的每个个体a代入到$P(x)$中的所有x，就对应一个可以判定真假意义的命题$P(a)$。不同的个体代入后所对应的命题真值可能不同，也可能相同。
- 例如,$P(x)$表示为
- $x^2-1=(x-1)(x+1)$　　x指定的个体域为全体整数，

- 则对任意整数i,
- $i^2-1=(i-1)(i+1)$
- 恒成立。也就是说，该命题函数的真值无论用什么个体代入总是对应为T。此类命题函数的真值描述通过一个称为全称量词的特殊符号来量化。

- 定义2 命题函数P(x)的全称量化（universal quanification）是一个按如下规则确定真值的命题：如果对每一个个体a代入得到的P(a)均为T。则该命题为T。记为$\forall x P(x)$。这里$\forall$是全称量词（universal quantifier），表示为“对任意的”、“所有的”、“对每一个”等等。

DEFINITION 2.

- The universal quantification of P(x) is the proposition
- "P(x) is true for all values of x in the universe of discourse."

EXAMPLE 5

- Express the statement
- "Every student in this class has studied calculus"
- as a universal quantification.

It can be written as

$\forall x P(x)$ or $\forall x\,(S(x) \rightarrow P(x))$

P(x) = "x has studied calculus."

S(x) = "x is in this class."

EXAMPLE 8

- What is the truth value of $\forall x P(x)$, where P(x) is the statement "$x^2 < 10$" and the universe of discourse consists of the positive integers not exceeding 4?

$$\forall x\, P(x) = P(1) \wedge P(2) \wedge P(3) \wedge P(4) = .F.$$

- 定义3　命题函数P(x)的存在量化（existential quantification）是一个按如下规则确定真值的命题：如果个体域中存在一个个体a代入后得到P(a)为T，则该命题为T，否则为F。记∃或э 。这里∃称为存在量词（existential quantification）。表示为“有一个”、“某些”、“某个”等等。

DEFINITION 3.

- The existential quantification of P(x) is the proposition
- “There exists an element x in the universe of discourse such that P(x) is true.”

EXAMPLE 9

- Let P(x) denote the statement "x > 3." What is the truth value of the quantification $\exists x P(x)$, where the universe of discourse is the set of real numbers?

Since "x>3" is true, for instance, when x=4 the existential quantification of P(x), which is ∃ xP(x) is true.

EXAMPLE 10

- Let Q(x) denote the statement "x = x + 1." What is the truth value of the quantification $\exists x Q(x)$, where the universe of discourse is the set of real numbers?

.F.

EXAMPLE 11

- What is the truth value of $\exists x \; P(x)$, where P(x) is the statement "$x^2 > 10$" and the universe of discourse consists of the positive integers not exceeding 4?

$\exists$ x P(x)

=P(1) $\vee$P(2)$\vee$P(3)$\vee$P(4)

= .T.

Table1

TABLE 1 Quantifiers.		
Statement	When True?	When False?
$\forall x P(x)$ $\exists x P(x)$	P(x) is true for every x. There is an x for which P(x) is true.	There is an x for which P(x) is false. P(x) is false for every x.

Uniqueness Quantifier (*optional*)

- $\exists!x\ P(x)$ means that $P(x)$ is true for one and only one x in the universe of discourse.
- This is commonly expressed in English in the following equivalent ways:
  - "There is a unique x such that $P(x)$."
  - "There is one and only one x such that $P(x)$"
- Examples:
  1. If *P(x)* denotes "$x + 1 = 0$" and U is the integers, then $\exists!x\ P(x)$ is true.
  2. But if *P(x)* denotes "$x > 0$," then $\exists!x\ P(x)$ is false.
- The uniqueness quantifier is not really needed as the restriction that there is a unique *x* such that *P*(*x*) can be expressed as:

$$\exists x\ (P(x) \wedge \forall y\ (P(y) \rightarrow y = x))$$

- 定义4　谓词公式定义为
- （1）n元谓词是一个谓词公式；
- （2）若A是谓词公式，则（¬A）也是谓词公式；
- （3）若A，B是谓词公式，则（A∨B）、（A∧B）、（A→B）、（A ↔ B）也是谓词公式；
- （4）若A是谓词公式且含有未被量化的个体变量x，则
- ∀xA(x)，∃XA(x)也是谓词公式。
- （5）有限次地使用（1）～（4）所得到的也是谓词公式。

$$\forall x ((F(x) \wedge P(x)) \rightarrow \exists y M(x, y)).$$

- 变量的约束（BINDING VARIABLES）
- 1,自由变量（Free Variable）
- 2,约束变量（Bound Variable）

More to Know About Binding



- $(\forall x\, P(x)) \wedge Q(x)$ - The variable x is outside of the *scope* of the $\forall x$ quantifier, and is therefore free. Not a complete proposition!





- $(\forall x\, P(x)) \wedge (\exists x\, Q(x))$ – A complete proposition, and no superfluous quantifiers

变元的约束

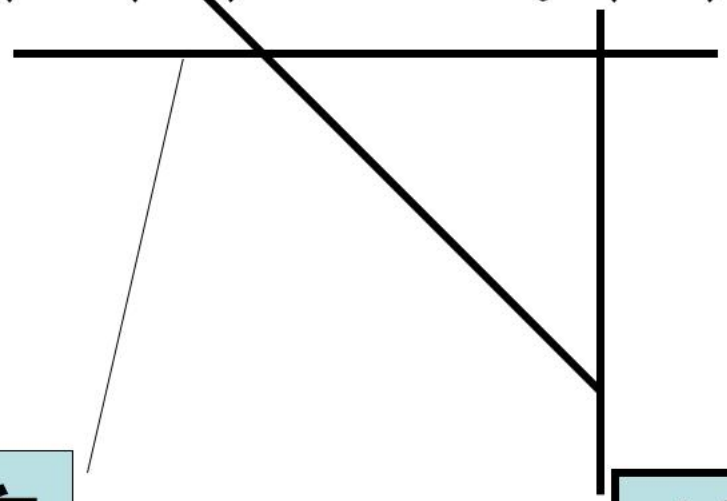
在谓词公式中,形如 $\forall xP(x), \exists xP(x)$ 的部分,称为谓词公式的 约束部分.$\forall, \exists$ 后跟的 X 叫量词的指导变元或作用变元，$P(x)$ 叫做相应量词的作用域或辖域.scope

在作用域中X 的一切出现，称为X 在公式中的约束出现. 所有约束出现的变元叫做约束变元.

不受约束的变元为自由变元.

例

1) $\forall x(P(x) \rightarrow Q(x))$

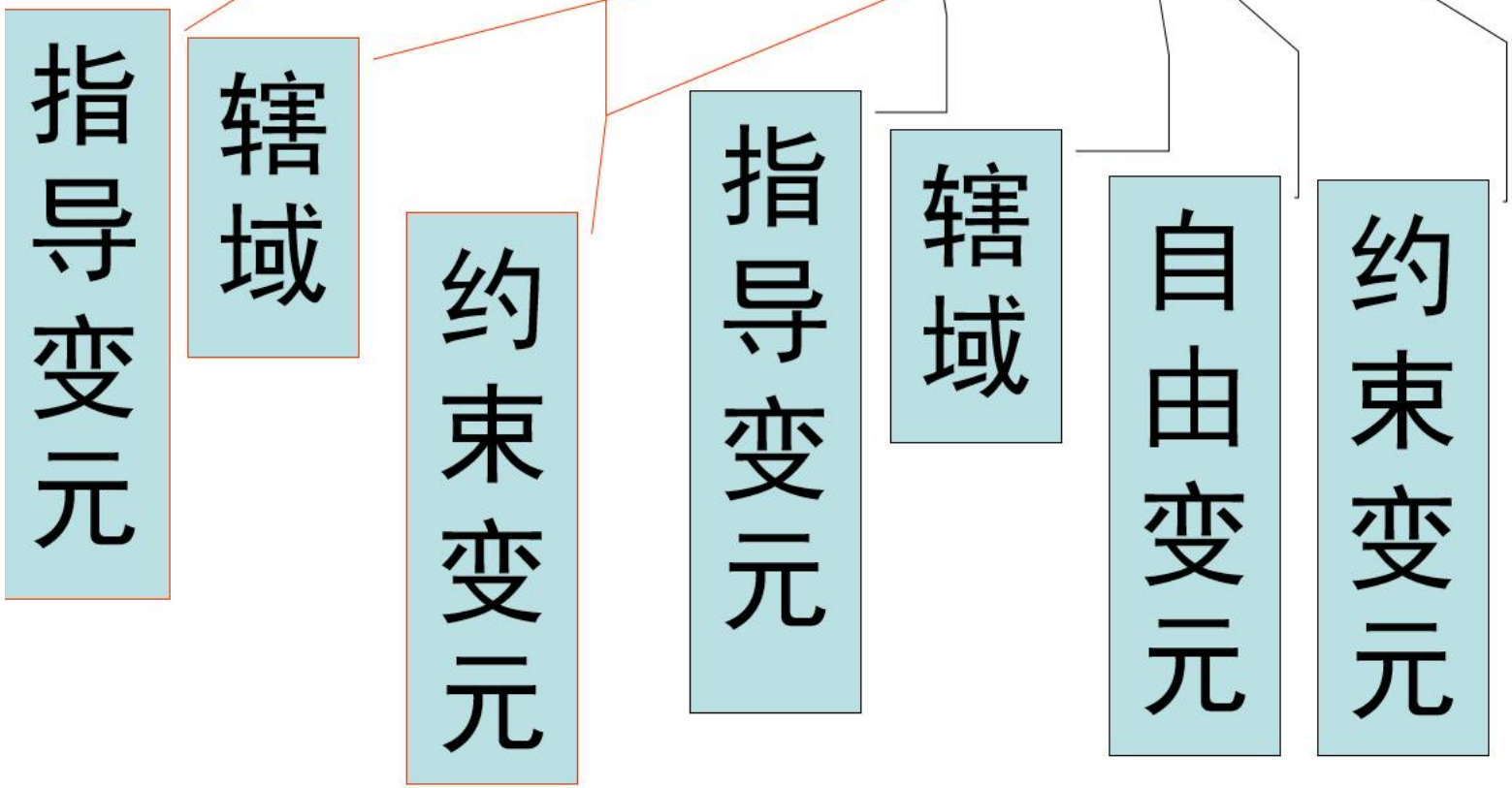


指导变元

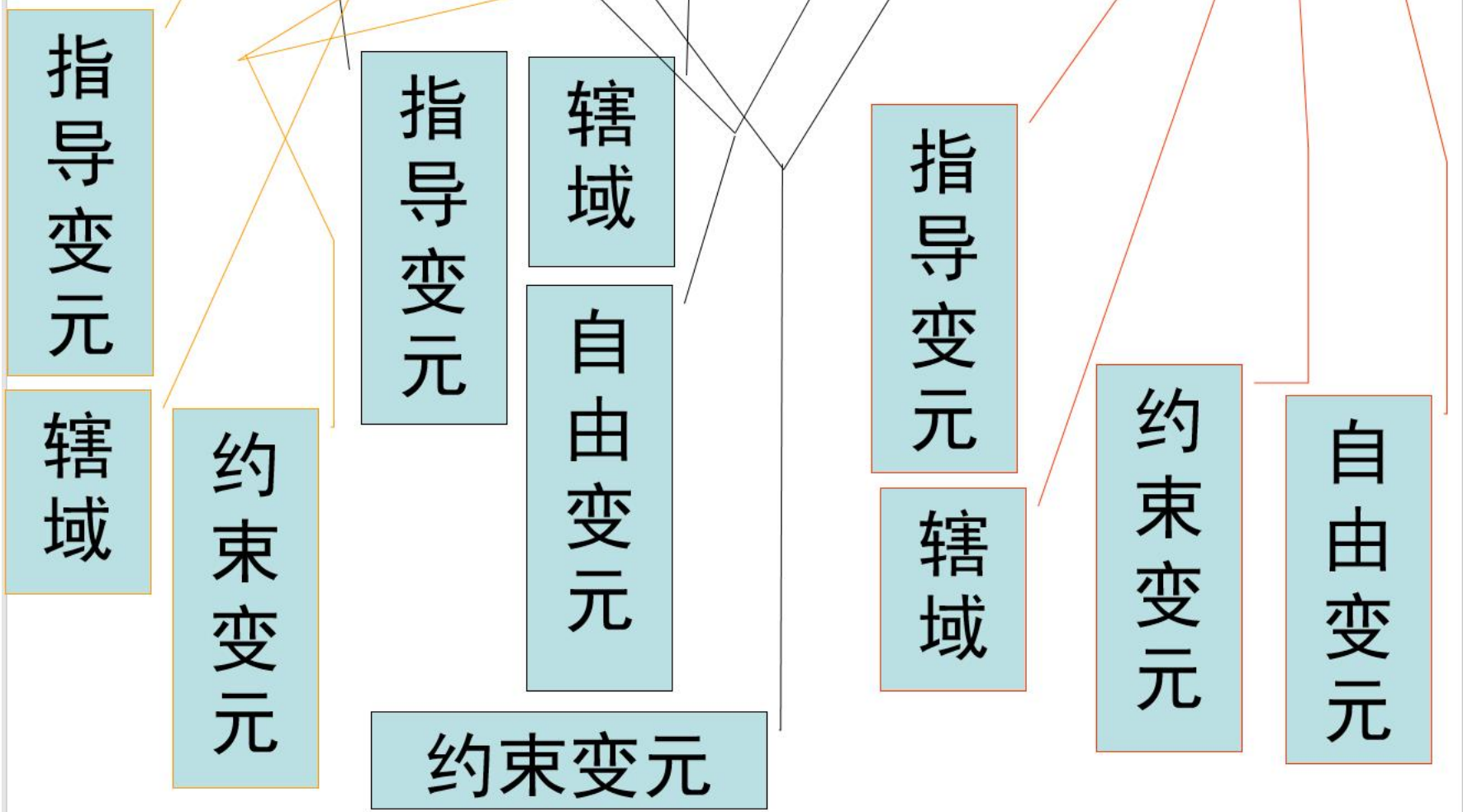
辖域

约束变元

2) $\forall x\,(P(x) \rightarrow \exists y\, R(x, y))$



3) $\forall x \forall y (P(x, y) \wedge Q(x, y)) \wedge \exists x P(x, y)$



P(x_1,…,x_n)是n 元谓词，有n个独立的自由变元。若对其中K个变元进行约束，则成为n-k元谓词。若没有自由变量出现，则该式变成为有关命题。

谓词公式的翻译
解释(interpret)

- 对一个合式公式的解释包括给出
- 个体域
- 谓词
- 函数
- 个体常项

的具体含义

EXAMPLE 12

- Translate the statement
- $\forall x\,(C(x) \vee \exists\, y(C(y) \wedge F(x, y)))$
- **The statement says that for every student x in BUPT x has a computer or there is a student y such that y has a computer and x and y are friends. In other words, every student in BUPT has a computer or has a friend who has a computer.**

EXAMPLE 13

- Translate the statement
- $\exists x \forall y \forall z(((F(x,y) \wedge F(x,z) \wedge (y \neq z)) \rightarrow \neg F(y,z)))$
- into English, where F(a,b) means a and b

This statement says that there is a student x such that for all students y and all students z other than y, if x and y are friends and x and z are friends, then y and z are not friends. In other words, there is a student none of whose friends are also friends with each other.

$$\forall x \forall y(P(x,y) \rightarrow Q(y))$$

- 个体域：　大于0的整数
- P(x,y):　y>x
- Q(y):　y>1

- 个体域：　整数
- P(x,y):　y>x
- Q(y):　y>1

语句形式化过程的三个关键点

- 准确地从语句中提取谓词，表示性质的谓语用一元谓词，表示关系的谓词用二元或更多元数的谓词来表示。
- 准确地使用量词和确定量词的辖域，当辖域中多于一个谓词时必须注意联结词与括号的使用。全称蕴涵　存在合取
- 准确地使用多个重叠的量词，其排列次序应与原语句意义一致。

Example 23

- Express the statement “ Every student in this class has studied calculus” using predicates and quantifiers.

- $\forall x\,(S(x) \rightarrow C(x))$
- 区别:
- $\forall x\,(S(x) \vee E(x))$

EXAMPLE 24

- Express the statements "Some student in this class has visited Mexico" and "Every student in this class has visited either Canada or Mexico" using quantifiers.

- $\exists x\,(S(x) \wedge M(x))$
- 区别:
- $\exists x\,(S(x) \rightarrow M(x))$

$\forall x(C(x) \vee M(x))$

$\forall x\,(S(x) \rightarrow (C(x) \vee M(x))$

Example 25

- Use predicates and quantifiers to express the system specifications “Every mail message larger than one megabyte will be compressed” and “If a user is active, at least one network link will be available.”

- Let S(m,y) be “Mail message *m* is larger than y megabytes” where the variable x has the domain of all mail messages and the variable y is a positive real number, and let C(m) denote “Mail message *m* will be compressed.”
- $\forall$m (S(m,1) $\rightarrow$ C(m))
- A(u):User *u* is active where the variable u has the domain of all users, S(n,x): network link *n* is in state *x* where n has the domain of all network links and x has the domain of all possible states for a network link
- $\exists$uA(u) $\rightarrow$ $\exists$nS(n, *available*)

例1：每个人都爱她自己的孩子

- 令
  - $P(x)$：x是人
  - $C(x)$：x是孩子
  - $I(x，y)$：x属于y
  - $L(x，y)$：x爱y
- 一阶逻辑表示
  - $\forall x \forall y(P(y) \wedge C(x) \wedge I(x，y) \rightarrow L(y，x))$

例2

- 航海家都教育自己的孩子成为航海家，有一个人教育他的孩子去做飞行员，证明：这个人一定不是航海家。
- 令
  - $S(x)$：x是航海家
  - $E(x)$：x教育他的孩子成为航海家

例2

- 前提：
  - P_1：$\forall x\,(S(x) \rightarrow E(x))$
  - P_2：$\exists x\,(\neg E(x))$
- 结论：
  - C：$\exists x\,(\neg E(x) \wedge \neg S(x))$
- 于是，要证明：
  - $P_1 \wedge P_2 \rightarrow C$
  - $\forall x\,(S(x) \rightarrow E(x)) \wedge \exists x\,(\neg E(x)) \rightarrow \exists x\,(\neg E(x) \wedge \neg S(x))$
- 是永真公式

例3

- 假设有一个人被每一个喜欢某些人的人所喜欢，又假设没有不喜欢人的人。证明有一个人被所有的人喜欢。
- 令
  - $L(x，y)$：x喜欢y
- 前提：
  - P_1：$\exists y \forall x\,(\exists z\, L(z，y) \rightarrow L(x，y))$
  - P_2：$\neg\exists x \forall y\,(\neg L(x，y))$
- 结论：
  - C：$\exists x \forall y\, L(y，x)$
- 于是，要证明：
  - $P_1 \wedge P_2 \rightarrow C$

例3

- 前提：
  - P_1： $\exists y \forall x\,(\exists z\, L(z，y) \rightarrow L(x，y))$
  - P_2： $\neg\exists x \forall y\,(\neg L(x，y))$
- 结论：
  - C： $\exists x \forall y\, L(y，x)$
- 于是，要证明：
  - $P_1 \wedge P_2 \rightarrow C$
  - $\exists y \forall x\,(\exists z\, L(z，y) \rightarrow L(x，y)) \wedge \neg\exists x \forall y\,(\neg L(x，y)) \rightarrow \exists x \forall y\, L(y，x)$
- 是永真公式

例4

- 每一个人的外祖父都是他母亲的父亲
- 令
  - $P(x)$：x是人
  - $O(x，y)$：x是y的外祖父
  - $F(x，y)$：x是y的父亲
  - $M(x，y)$：x是y的母亲
- 命题表示
  - $\forall x \forall y\,(P(y) \wedge O(x，y) \rightarrow \exists z\,(F(x，z) \wedge M(z，y)))$

例5

- 不管黑猫白猫，抓住老鼠就是好猫
- 令
  - $C(x)$：x是猫
  - $B(x)$：x是黑的
  - $W(x)$：x是白的
  - $G(x)$：x是好的
  - $M(x)$：x是老鼠
  - $K(x，y)$：x抓住y
- 命题表示
  - $\forall x \forall y\,(C(x) \wedge M(y) \wedge (B(x) \vee W(x)) \wedge K(x，y) \rightarrow G(x))$

例6

- 有些病人相信所有的医生，但病人都不相信一个骗子。证明：医生都不是骗子
- 令
  - $F(x)$：x是病人
  - $G(x)$：x是医生
  - $H(x)$：x是骗子
  - $L(x，y)$：x相信y

例6

- 前提：
  - G_1：$\exists x\,(F(x) \wedge \forall y\,(G(y) \rightarrow L(x, y)))$
  - G_2：$\forall x \forall y\,(F(x) \wedge H(y) \rightarrow \neg L(x, y))$
- 结论：
  - C：$\forall x\,(G(x) \rightarrow \neg H(x))$
- 于是，要证明：
  - $G_1 \wedge G_2 \rightarrow C$
- 是永真公式

例7

- 每个有理数都是实数
- 某些实数是有理数
- 不是每一个实数都是有理数

例

并非每个实数都是有理数

$R(x)$：x 是实数

$Q(x)$：x 是有理数

$$\neg \forall x(R(x) \rightarrow Q(x))$$

$$\exists x(R(x) \wedge \neg Q(x))$$

例8

- 对平面上任意两点，有且仅有一条直线通过这两点
- 令
  - $P(x)$：x是一个点
  - $L(x)$：x是一条直线
  - $R(x，y，z)$：z通过x和y
  - $E(x，y)$：x等于y

例8

- $P(x)$：x是一个点
- $L(x)$：x是一条直线
- $R(x，y，z)$：z通过x和y
- $E(x，y)$：x等于y

- $\forall x \forall y\,(P(x) \wedge P(y) \rightarrow \exists z\,(L(z) \wedge R(x，y，z) \wedge \forall u\,(L(u) \wedge R(x，y，u) \rightarrow E(z，u))))$

例9

- 在实数集中，任给一正实数，都存在大于该实数的实数

例10：函数序列收敛

- 设$f_1(x)$, …, $f_n(x)$, …是函数序列，$f(x)$是一个函数。
- “对任给$\varepsilon>0$, $x_0 \in (a, b)$, 都存在N，使得当$n>N$时，有
  - $| f(x_0) - f_n(x_0) | < \varepsilon$
- 则称函数序列$\{f_n(x)\}$在(a, b)区间内收敛于$f(x)$”

例10：函数序列收敛

- 令
  - $G(x，y)$：x大于y
  - $B(x)$：x属于(a, b)区间
  - $s(x，n)$： $f(x)$与 $f_n(x)$ 之差的绝对值
- 于是
  - $\forall x \forall \varepsilon (G(\varepsilon，0) \wedge B(x) \rightarrow \exists N \forall n\ (G(n，N) \rightarrow G(\varepsilon，s(x，n))))$

例11：函数序列一致收敛

- 设$f_1(x)$, …, $f_n(x)$, …是函数序列，$f(x)$是一个函数。
- “对任给$\varepsilon>0$, 都存在N，使得当$n>N$时，对所有 $x\in(a, b)$，都有
  - $|f(x)-f_n(x)|<\varepsilon$
- 则称函数序列$\{f_n(x)\}$在(a, b)区间内一致收敛于$f(x)$”

例

没有不犯错误的人

M(X) X是人　F(X) X犯错误

$$\forall x\,(M(x) \rightarrow F(x))$$

$$\neg\exists\, x\,(M(x) \wedge \neg F(x))$$

例

发光的不都是金子

$L(x)$：x 发光

$G(x)$：x 是金子

$$\neg\forall x(L(x) \rightarrow G(x))$$

$$\exists x(L(x) \wedge \neg G(x))$$

例

不是所有男人都比女人高

M(x): x 是男人, W(x): x 是女人

H(x,y),x比y 高

$$\exists x(M(x) \wedge \exists y(W(y) \wedge H(y, x)))$$

$$\neg\forall\ x\ (M(x) \rightarrow \forall y\ (W(y) \rightarrow H(x, y)))$$

例

有人喜欢吃糖

$M(x)$：x 是人 $S(x)$：x 是糖

$L(x, y)$：x 喜欢

$$\exists x(M(x) \wedge \exists y(S(y) \wedge H(x, y)))$$

$$\neg\forall x(M(x) \rightarrow \forall y(S(y) \rightarrow L(x, y)))$$

- Can predicate logic say “there exist exactly two objects with property P”?

- Can predicate logic say “there exist exactly two objects with property P”?
- Yes:
- $\exists x \, \exists y \, (P(x) \wedge P(y) \wedge x \neq y \; \wedge$
  $\forall z \, (P(z) \rightarrow (z = x \vee z = y)))$

What's wrong with

- $\exists x\, \exists y\, (P(x) \wedge P(y) \wedge x \neq y) \;\wedge$
  $\forall z\, (P(z) \rightarrow (z = x \vee z = y))$
  as a formalisation of `exactly two'?

What's wrong with

- $\exists x\, \exists y\, (P(x) \wedge P(y) \wedge x \neq y) \;\wedge$
  $\forall z\, (P(z) \rightarrow (z = x \vee z = y))$
  as a formalisation of `exactly two'?
- This is a conjunction of two separate propositions. As a result, *x* and *y* are not bound, so this is not even a proposition

EXAMPLE 16

- Express the statement "Everyone has exactly one best friend" as a logical expression.

$$\forall x \exists y \forall z (B(x, y) \wedge ((z \neq y) \rightarrow \neg B(x,z))).$$

EXAMPLE 17

- Express the statement "If somebody is female and is a parent, then this person is someone's mother" as a logical expression.

$$\forall x ((F(x) \wedge P(x)) \rightarrow \exists y M(x, y)).$$

EXAMPLE 18

- Use quantifiers to express the statement "There is a woman who has taken a flight on every airline in the world."

P(w,f): w has taken a f.

Q(f,a): f is a flight on a.

$$\exists w \forall a \exists f(P(w, f) \wedge Q(f, a))$$

EXAMPLE 19

- (Calculus required) Express the definition of a limit using quantifiers.

$$\forall\ \varepsilon \exists\ \delta\ \forall\ x\,(\ 0 < |\ x - a\ | < \delta \rightarrow |\ f(x) - L\ | < \varepsilon$$

$$\forall\ \varepsilon > 0 \exists\ \delta\ > 0 \forall\ x\,(0 < |\ x - a\ | < \delta \rightarrow |\ f(x) - L\ | < \varepsilon$$

- Consider the following statements. The first two are called premises and conclusion. The entire se argument.

$$\forall x(P(x) \rightarrow Q(x)).$$
$$\exists x(P(x) \wedge \neg R(x)).$$
$$\exists x(Q(x) \wedge \neg R(x)).$$

- **"All lions are fierce."**
- **"Some lions do not drink coffee."**
- **"Some fierce creatures do not drink coffee."**
- **Let P(x), Q(x), and R(x) be the statements "x is a lion," "x is fierce," and "x drinks coffee," respectively. Assuming that the universe of discourse is the set of all creatures, express the statements in the argument using quantifiers and P(x), Q(x), and R(x).**

- Consider the following statements, of which the first three are premises and the four conclusion.

$\forall x(P(x) \rightarrow S(x))$.
$\neg \exists x(Q(x) \wedge R(x))$.
$\forall x(\neg R(x) \rightarrow \neg S(x))$.
$\forall x(P(x) \rightarrow \neg Q(x))$.

- **"All hummingbirds are richly co**
- **"No large birds live on honey."**
- **"Birds that do not live on honey are dull in color."**
- **"Hummingbirds are small."**
- **Let P(x), Q(x), R(x), and S(x) be the statements "x is a hummingbird," "x is large," "x lives on honey," and "x is richly colored," respectively. Assuming that the universe of discourse is the set of all birds, express the statements in the argument using quantifiers and P(x), Q(x), R(x), and S(x).**

EXAMPLE 22

- Let Q(x, y) denote "x + y = 0." What are the truth values of the quantifications

- $\exists y \forall x Q(x, y)$ and $\forall x \exists y Q(x, y)$?

"There is a real number y such that for every real number x, Q(x, y) is true."

No matter what value of y is chosen, there is only one value of x for which x + y = 0. Since there is no real number y such that x + y = 0 for all real numbers x, **the statement is false.**

"For every real number x there is a real number y such that Q(x, y) is true."

Given a real number x, there is a real number y such that x + y = 0; namely, y = −x. **Hence, the statement is true.**

Logic Programming

- An important type of programming language is designed to reason using the rules of predicate logic. Prolog (from Programming in Logic), developed in the 1970s by computer scientists working in the area of artificial intelligence, is an example of such a language.
- Prolog programs include a set of declarations consisting of two types of statements, Prolog facts and Prolog Links rules. Prolog facts define predicates by specifying the elements that satisfy these predicates.

1.5 Nested Quantifiers 嵌套量词

- Translating
- Negation
- The order

- Let Q(x, y, z) be the statement "x + y = z." What are the truth values of the statements
- $\forall x \forall y \exists z Q(x, y, z)$ and $\exists z \forall x \forall y Q(x, y, z)$?

"For all real numbers x and for all real numbers y there is a real number z such that x + y = z,"

the statement is true.

"There is a real number z such that for all real numbers x and for all real numbers y it is true that x + y = z,"

the statement is false.

Table1

TABLE 2 Quantifications of Two Variables.		
Statement	**When True?**	**When False?**
$\forall x \forall y P(x,y)$ $\forall y \forall x P(x,y)$	P(x, y) is true for every pair x, y.	There is a pair x, y for which P(x, y) is false.
$\forall x \exists y P(x,y)$	For every x there is a y for which P(x, y) is true.	There is an x such that P(x, y) is false for every y.
$\exists x \forall y P(x,y)$	There is an x for which P(x, y) is true for every y.	For every x there is a y for which P(x, y) is false.
$\exists x \exists y P(x,y)$ $\exists y \exists x P(x,y)$	There is a pair x, y for which P(x, y) is true.	P(x, y) is false for every pair x, y.

NEGATIONS

$$\neg \forall x P(x) \Leftrightarrow \exists x \neg P(x).$$

$$\neg \exists x Q(x) \Leftrightarrow \forall x \neg Q(x).$$

Table

TABLE 3 Negating Quantifiers.			
Negation	Equivalent Statement	When Is Negation True?	When False?
$\neg \exists x P(x)$ $\neg \forall x P(x)$	$\forall x \neg P(x)$ $\exists x \neg P(x)$	$P(x)$ is false for every x. There is an x for which $P(x)$ is false.	There is an x for which $P(x)$ is true. $P(x)$ is true for every x.

Nesting of Quantifiers

Example: Let the u.d. of x and y be people.

Let $L(x,y)$="x likes y" (a predicate w. 2 f.v.'s)

Then $\exists y\, L(x,y)$ = "There is someone whom x likes." (A predicate w. 1 free variable, x)

Then $\forall x\, (\exists y\, L(x,y))$ =
"Everyone has someone whom they like."

(a real proposition; no free variables left)

Quantifier Equivalence Laws

- Expanding quantifiers: If u.d.=a,b,c,…
  $\forall x\, P(x) \Leftrightarrow P(\mathrm{a}) \wedge P(\mathrm{b}) \wedge P(\mathrm{c}) \wedge \ldots$
  $\exists x\, P(x) \Leftrightarrow P(\mathrm{a}) \vee P(\mathrm{b}) \vee P(\mathrm{c}) \vee \ldots$
- From those, we can prove the laws:
  $\forall x\, P(x) \Leftrightarrow \neg\exists x\, \neg P(x)$
  $\exists x\, P(x) \Leftrightarrow \neg\forall x\, \neg P(x)$
- Which *propositional* equivalence laws can be used to prove this?

DeMorgan's

Negating nested quantifiers

- Example 12 There does not exit a woman who has taken a flight on every airline in the world.

$$\exists \mathrm{w} \forall \mathrm{a} \exists \mathrm{f}(\mathrm{P}(\mathrm{w}, \mathrm{f}) \wedge \mathrm{Q}(\mathrm{f}, \mathrm{a}))$$

$$\neg\exists w \forall a \exists f\,(P(w, f) \wedge Q(f, a))$$

$$\Leftrightarrow \forall w \neg \forall a \exists f\,(P(w, f) \wedge Q(f, a))$$

$$\Leftrightarrow \forall w \exists a \neg \exists f\,(P(w, f) \wedge Q(f, a))$$

$$\Leftrightarrow \forall w \exists a \forall f \neg (P(w, f) \wedge Q(f, a))$$

$$\Leftrightarrow \forall w \exists a \forall f\,(\neg P(w, f) \vee \neg Q(f, a))$$

Remember

- In propositional logic, we can *strictly speaking* only build formulas of finite size.
- E.g., we can write
  $P(a) \wedge P(b)$
  $P(a) \wedge P(b) \wedge P(c)$
  $P(a) \wedge P(b) \wedge P(c) \wedge P(d)$, etc.
- But this way, we could never say that all natural numbers have P

- In predicate logic, you can say this easily:
  $\forall x P(x)$
- Still, it's sometimes useful to pretend that propositional logic allows infinitely long formulas.

More Equivalence Laws

- $\forall x\ \forall y\ P(x,y) \Leftrightarrow \forall y\ \forall x\ P(x,y)$
  $\exists x\ \exists y\ P(x,y) \Leftrightarrow \exists y\ \exists x\ P(x,y)$
- $\forall x\ (P(x) \wedge Q(x)) \Leftrightarrow (\forall x\ P(x)) \wedge (\forall x\ Q(x))$
  $\exists x\ (P(x) \vee Q(x)) \Leftrightarrow (\exists x\ P(x)) \vee (\exists x\ Q(x))$
- How about this one?

  $\exists x\ (P(x) \wedge Q(x)) \Leftrightarrow (\exists x\ P(x)) \wedge (\exists x\ Q(x))$

More Equivalence Laws

- How about this one?

$$\exists x\,(P(x) \wedge Q(x)) \Leftrightarrow (\exists x\,P(x)) \wedge (\exists x\,Q(x))\ ?$$

- *This equivalence statement is false.*
  Counterexample:
  P(x): x's birthday is on 30 April
  Q(x): x's birthday is on 20 December

一阶逻辑等值式(来源)

- 命题逻辑等值式的代换实例
- 与量词有关的
  - 有限个体域量词消去
  - 量词否定
  - 量词辖域收缩与扩张
  - 量词分配
- 与变项命名有关的
  - 换名规则
  - 代替规则

代换实例

在命题逻辑等值式中, 代入一阶逻辑公式所得到的式子, 称为原来公式的代换实例.

例1：$A \Leftrightarrow \neg\neg A$, 令$A=\forall x F(x)$, 得到

$$\forall x F(x) \Leftrightarrow \neg\neg \forall x F(x)$$

例2：$A \rightarrow B \Leftrightarrow \neg A \vee B$, 令$A=F(x)$, $B=G(y)$, 得到

$$F(x) \rightarrow G(y) \Leftrightarrow \neg F(x) \vee G(y)$$

有限个体域上消去量词

设个体域为有限集$D=\{a_1, a_2,\ldots, a_n\}$, 则

$\forall xA(x)\Leftrightarrow A(a_1)\wedge A(a_2)\wedge\ldots\wedge A(a_n)$

$\exists xA(x)\Leftrightarrow A(a_1)\vee A(a_2)\vee\ldots\vee A(a_n)$

例: 个体域$D=\{a,b,c\}$, 则 $\exists x\forall yF(x,y)$

$\Leftrightarrow\exists x\,(F(x,a)\wedge F(x,b)\wedge F(x,c))$

$\Leftrightarrow\ (F(a,a)\wedge F(a,b)\wedge F(a,c))\vee$

$(F(b,a)\wedge F(b,b)\wedge F(b,c))\vee$

$(F(c,a)\wedge F(c,b)\wedge F(c,c))$

量词否定等值式

$\neg\ \forall x A(x) \Leftrightarrow \exists x\ \neg\ A(x)$

$\neg\ \exists x A(x) \Leftrightarrow \forall x\ \neg\ A(x)$

量词辖域收缩与扩张(∀)

- $\forall x(A(x) \vee B) \Leftrightarrow \forall x A(x) \vee B$
- $\forall x(A(x) \wedge B) \Leftrightarrow \forall x A(x) \wedge B$
- $\forall x(A(x) \rightarrow B) \Leftrightarrow \exists x A(x) \rightarrow B$
- $\forall x(B \rightarrow A(x)) \Leftrightarrow B \rightarrow \forall x A(x)$
- 说明： B中不含x的出现
- 例1: $\forall x(F(x) \vee G(y)) \Leftrightarrow \forall x F(x) \vee G(y)$
- 例2: $\forall x \forall y(F(x) \wedge G(y)) \Leftrightarrow \forall x(F(x) \wedge \forall y G(y))$
  $\Leftrightarrow \forall x F(x) \wedge \forall y G(y)$

量词辖域收缩与扩张(∀、续)

- $\forall x(A(x)\rightarrow B) \Leftrightarrow \exists xA(x)\rightarrow B$

  证明： $\forall x(A(x)\rightarrow B)$

  $\Leftrightarrow \forall x(\neg A(x)\vee B) \Leftrightarrow \forall x \neg A(x)\vee B$

  $\Leftrightarrow \neg \exists xA(x)\vee B \Leftrightarrow \exists xA(x)\rightarrow B$

- $\forall x(B\rightarrow A(x)) \Leftrightarrow B\rightarrow \forall xA(x)$

  证明： $\forall x(B\rightarrow A(x))$

  $\Leftrightarrow \forall x(\neg B\vee A(x))$

  $\Leftrightarrow \neg B\vee \forall xA(x) \Leftrightarrow B\rightarrow \forall xA(x)$

量词辖域收缩与扩张(∃)

- $\exists x(A(x) \vee B) \Leftrightarrow \exists x A(x) \vee B$
- $\exists x(A(x) \wedge B) \Leftrightarrow \exists x A(x) \wedge B$
- $\exists x(A(x) \rightarrow B) \Leftrightarrow \forall x A(x) \rightarrow B$
- $\exists x(B \rightarrow A(x)) \Leftrightarrow B \rightarrow \exists x A(x)$
- 说明： B中不含x的出现
- 例1: $\exists x(F(x) \vee G(y)) \Leftrightarrow \exists x F(x) \vee G(y)$
- 例2: $\forall x \exists y(F(x) \wedge G(y)) \Leftrightarrow \forall x(F(x) \wedge \exists y G(y))$
  $\Leftrightarrow \mathbf{\forall x F(x) \wedge \exists y G(y)}$

量词辖域收缩与扩张(∃、续)

- $\exists x(A(x)\rightarrow B) \Leftrightarrow \forall xA(x)\rightarrow B$

  证明：$\exists x(A(x)\rightarrow B)$

  $\Leftrightarrow \exists x(\neg A(x)\vee B) \Leftrightarrow \exists x \neg A(x)\vee B$

  $\Leftrightarrow \neg \forall xA(x)\vee B \Leftrightarrow \forall xA(x)\rightarrow B$

- $\exists x(B\rightarrow A(x)) \Leftrightarrow B\rightarrow \exists xA(x)$

  证明：$\exists x(B\rightarrow A(x))$

  $\Leftrightarrow \exists x(\neg B\vee A(x)) \Leftrightarrow \neg B\vee \exists xA(x)$

  $\Leftrightarrow \neg B\vee \exists xA(x) \Leftrightarrow B\rightarrow \exists xA(x)$

量词分配

- $\forall x(A(x) \wedge B(x)) \Leftrightarrow \forall xA(x) \wedge \forall xB(x)$
- $\exists x(A(x) \vee B(x)) \Leftrightarrow \exists xA(x) \vee \exists xB(x)$

量词分配(反例)

- $\forall x(A(x) \vee B(x)) \Leftarrow \forall xA(x) \vee \forall xB(x)$
- $\forall x(A(x) \vee B(x)) \Rightarrow \forall xA(x) \vee \forall xB(x)$

个体域为全体自然数; A(x): x是偶数

B(x): x是奇数;　左⇔1, 右⇔0

- $\exists x(A(x) \wedge B(x)) \Rightarrow \exists xA(x) \wedge \exists xB(x)$
- $\exists x(A(x) \wedge B(x)) \Leftarrow \exists xA(x) \wedge \exists xB(x)$

个体域为全体自然数; A(x): x是偶数

B(x): x是奇数;　左⇔0, 右⇔1

换名(rename)规则

- 把某个指导变项和其量词辖域中所有同名的约束出现, 都换成某个新的个体变项符号.
- 例如:
- $\forall x(A(x) \wedge B(x)) \Leftrightarrow \forall y(A(y) \wedge B(y))$
- $\forall x A(x) \wedge \forall x B(x) \Leftrightarrow \forall y A(y) \wedge \forall z B(z)$
- $H(x,y) \vee \exists x F(x) \vee \forall y(G(y) \rightarrow H(x,y))$

$\Leftrightarrow H(x,y) \vee \exists z F(z) \vee \forall u(G(u) \rightarrow H(x,u))$

代替(substitute)规则

- 把某个自由变项的所有出现, 都换成某个新的个体变项符号.
- 例如:
- $A(x) \wedge B(x) \Leftrightarrow A(y) \wedge B(y)$
- $\forall x A(x) \wedge B(x) \Leftrightarrow \forall x A(x) \wedge B(y)$
- $H(x,y) \vee \exists x F(x) \vee \forall y (G(y) \rightarrow H(x,y))$

$\Leftrightarrow H(s,t) \vee \exists x F(x) \vee \forall y (G(y) \rightarrow H(s,y))$

Dangerous situations

- Commutativity of quantifiers
  - $\forall x \forall y P(x, y) \Leftrightarrow \forall y \forall x P(x, y)$?

    YES!
  - $\forall x \exists y P(x, y) \Leftrightarrow \exists y \forall x P(x, y)$?

    NO!
  - DIFFERENT MEANING!

5) 蕴含式

$$\forall xA(x) \vee \forall xB(x)$$

$$\Rightarrow \forall x(A(x) \vee B(x))$$

$$\exists x(A(x) \wedge B(x))$$

$$\Rightarrow \exists xA(x) \wedge \exists xB(x)$$

$$\forall x(A(x) \leftrightarrow B(x))$$

$$\Rightarrow \forall xA(x) \leftrightarrow \forall xB(x)$$

6) 含多个变元的等价式和蕴含式

$$\forall x\forall yP(x, y) \Leftrightarrow \forall y\forall xP(x, y)$$

$$\forall x\forall yP(x, y) \Rightarrow \exists y\forall xP(x, y)$$

$$\forall y \forall x P(x, y) \Rightarrow \exists x \forall y P(x, y)$$

$$\exists y \forall x P(x, y) \Rightarrow \forall x \exists y P(x, y)$$

$$\exists x \forall y P(x, y) \Rightarrow \forall y \exists x P(x, y)$$

$$\forall x \exists y P(x, y) \Rightarrow \exists y \exists x P(x, y)$$

$$\forall y \exists x P(x, y) \Rightarrow \exists x \exists y P(x, y)$$

$$\exists x \exists y P(x, y) \Leftrightarrow \exists y \exists x P(x, y)$$

$\forall x \forall y$
$\forall y \forall x$
$\exists x \forall y$
$\exists y \forall x$
$\forall x \exists y$
$\forall y \exists x$
$\exists y \exists x$
$\exists x \exists y$

谓词公式的范式

- 前束范式prenex normal forms：一个公式，如果量词均在全式的开头，它们的作用域，延伸到整个公式的末尾，则该公式称为前束范式。
- 如：
- $(\forall x)(\forall y)(\exists z)(Q(x,y) \rightarrow R(z))$
- $(\forall y)(\forall x)(\neg P(x,y) \rightarrow Q(y))$
- 方法：将¬后移，辖域扩张或收缩，换名和替换规则等值演算

任何一个谓词公式均有与其等价的前束范式。

任何一个谓词公式均可转为与其等价的前束合取范式。

任何一个谓词公式均可转为与其等价的前束析取范式。

求前束范式

$$1)\ \forall xF(x) \wedge \neg\exists xG(x)$$

$$\Leftrightarrow\ \forall xF(x) \wedge \forall x\neg G(x)$$

$$\Leftrightarrow\ \forall x(F(x) \wedge \neg G(x))$$

$$2)\ \forall xF(x) \vee \neg\exists xG(x)$$

$$\Leftrightarrow\ \forall xF(x) \vee \forall x\neg G(x)$$

$$\Leftrightarrow \quad \forall x F(x) \vee \forall y \neg G(y)$$

$$\Leftrightarrow \quad \forall x \forall y (F(x) \vee \neg G(y))$$

$$3) \; (\forall x F(x, y) \rightarrow \exists y G(y))$$

$$\rightarrow \forall x H(x, y)$$

$$\Leftrightarrow \quad (\forall x F(x, z) \rightarrow \exists y G(y))$$

$$\rightarrow \forall x H(x, z)$$

$$\Leftrightarrow\ (\forall\, xF(x,\, z)\ \rightarrow\ \exists\, yG(y))$$

$$\rightarrow\ \forall\, t\ H(t,\, z)$$

$$\Leftrightarrow\ \exists\, x(F(x,\, z)\ \rightarrow\ \exists\, yG(y))$$

$$\rightarrow\ \forall\, t\ H(t,\, z)$$

$$\Leftrightarrow\ (\exists\, x\exists\, y(F(x,\, z)\ \rightarrow\ G(y)))$$

$$\rightarrow\ \forall\, t\ H(t,\, z)$$

$$\Leftrightarrow \ \forall x \forall y ((F(x, z) \ \rightarrow \ G(y))$$

$$\rightarrow \ \forall t \ H(t, z)))$$

$$\Leftrightarrow \ \forall x \forall y \forall t ((F(x, z) \ \rightarrow \ G(y))$$

$$\rightarrow H(t, z)))$$

$$4)\ \exists x F(x) \wedge \exists x G(x)$$

$$\Leftrightarrow\ \exists x F(x) \wedge \exists y G(y)$$

$$\Leftrightarrow\ \exists x (F(x) \wedge \exists y G(y))$$

$$\Leftrightarrow\ \exists x \exists y (F(x) \wedge G(y))$$

例1 把 $\forall xP(x) \rightarrow \exists xQ(x)$ 化为前束范式。

解:

原式 $\Leftrightarrow \neg\forall xP(x) \vee \exists xQ(x)$

$\Leftrightarrow \exists x\neg P(x) \vee \exists xQ(x)$

$\Leftrightarrow \exists x(\neg P(x) \vee Q(x))$

$\Leftrightarrow \exists x(P(x) \rightarrow Q(x))$

例2 把

$$\forall x \forall y (\exists z (P(x, z) \wedge P(y, z)) \rightarrow \exists u Q(x, y, u))$$

化为前束范式。

解: 原式 $\Leftrightarrow$

$$\forall x \forall y \forall z (P(x, z) \wedge P(y, z) \rightarrow \exists u Q(x, y, u))$$

$$\Leftrightarrow$$

$$\forall x \forall y \forall z \exists u (P(x, z) \wedge P(y, z)$$

$$\rightarrow Q(x, y, u))$$

Put these statements in prenex normal form.

- $\exists x S(x) \rightarrow \exists x T(x)$

Goldbach's Conjecture (unproven)

Using $E(x)$ and $P(x)$ from previous slide,

$$\forall x([x>2 \wedge E(x)] \rightarrow \exists p \exists q\ P(p) \wedge P(q) \wedge p+q = x).$$

"Every even number greater than 2 is the sum of two primes."

Calculus Example

- One way of precisely defining the calculus concept of a *limit*, using quantifiers:

$$\left(\lim_{x \to a} f(x) = L\right) \Leftrightarrow$$

$$\left(\begin{array}{l} \forall \varepsilon > 0 : \exists \delta > 0 : \forall x : \\ \left(|\, x - a \,| < \delta\right) \rightarrow \left(|\, f(x) - L \,| < \varepsilon\right) \end{array}\right)$$

End of § 1.4-1.5, Predicate Logic

- From these sections you should have learned:
  - Predicate logic notation & conventions
  - Conversions: predicate logic $\leftrightarrow$ clear English
  - Meaning of quantifiers, equivalences
  - Simple reasoning with quantifiers
- Upcoming topics:
  - Introduction to proof-writing.
  - Then: Set theory –
    - a language for talking about collections of objects.

作业

- § 1.4 – 10, 18, 30, 42,56, 62
- § 1.5 – 10, 28, 34, 46,50

1 The Foundations: Logic and Proofs

- **1.1** Propositional Logic
  1.2 Propositional Equivalences
  1.3 Predicates and Quantifiers
  1.4 Nested Quantifiers
  1.5 Rules of Inference
  1.6 Introduction to Proofs
  1.7 Proof Methods and Strategy

1 The Foundations: Logic and Proofs

- **1.1** Propositional Logic
- 1.2Applications of Propositional Logic
  1.3 Propositional Equivalences
  1.4 Predicates and Quantifiers
  1.5 Nested Quantifiers
  1.6 Rules of Inference
  1.7 Introduction to Proofs
  1.8 Proof Methods and Strategy

Rules of Inference

Rosen 8th ed., § 1.6

Argument(论证)

- A sequence of statements that end with a conclusion.
- “If you have a current password, then you can log onto the network.”
- “You have a current password.”
- Therefore
- “You can log onto the network.”

Definition 1

- An *argument* in propositional logic is a sequence of propositions. All but the final proposition in the argument are called ***premises*** and the final proposition is called the *conclusion*. An argument is ***valid*** if the truth of all its premises implies that the conclusion is true.

Argument form

- An *argument form* in propositional logic is a sequence of compound propositions involving propositional variables. An argument form is *valid* if no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if the premises are all true.
- Corresponding tautology:

$((p_1) \wedge (p_2) \wedge \ldots \wedge (p_n)) \rightarrow q$

Nature & Importance of Proofs

- In mathematics, a *proof* is:
  - a *correct* (well-reasoned, logically valid) and *complete* (clear, detailed) argument that rigorously & undeniably establishes the truth of a mathematical statement.
- Why must the argument be correct & complete?
  - *Correctness* prevents us from fooling ourselves.
  - *Completeness* allows anyone to verify the result.
- In this course (& throughout mathematics), a very high standard for correctness and completeness of proofs is demanded!!

Overview of § 1.6

- Methods of mathematical argument (*i.e.*, proof methods) can be formalized in terms of *rules of logical inference*.
- Mathematical *proofs* can themselves be represented formally as discrete structures.
- We will review both correct & fallacious inference rules, & several proof methods.

Inference Rules - General Form

- An *Inference Rule* is
  - A pattern establishing that if we know that a set of *antecedent* statements of certain forms are all true, then we can validly deduce that a certain related *consequent* statement is true.
- *antecedent 1*
  antecedent 2 ...

  ∴ *consequent*
  "therefore"

"∴" means

Inference Rules & Implications

- Each valid logical inference rule corresponds to an implication that is a tautology.
- $\dfrac{\textit{antecedent 1}\quad \textit{antecedent 2}\ \ldots}{\therefore\ \textit{consequent}}$ Inference rule
- Corresponding tautology:

  $((\textit{ante. 1}) \wedge (\textit{ante. 2}) \wedge \ldots) \rightarrow \textit{consequent}$

Some Inference Rules

- p / $\therefore p \vee q$ — Rule of Addition
- $p \wedge q$ / $\therefore p$ — Rule of Simplification
- p, q / $\therefore p \wedge q$ — Rule of Conjunction

Modus Ponens & Tollens

- p
  $p \rightarrow q$
  $\therefore q$

  Rule of *modus ponens*假言推理
  (a.k.a. *law of detachment*)

  "the mode of affirming"

- $\neg q$
  $p \rightarrow q$
  $\therefore \neg p$

  Rule of *modus tollens*拒取式

  "the mode of denying"

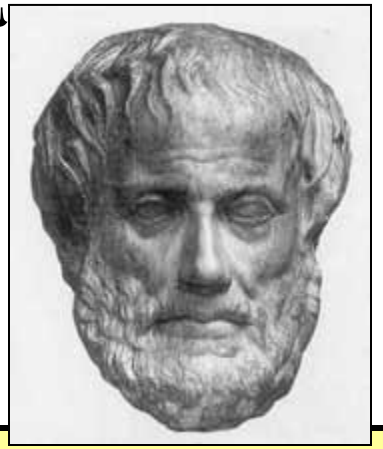
Syllogism Inference Rules

- $p \rightarrow q$
  $q \rightarrow r$
  $\therefore p \rightarrow r$

  Rule of hypothetical syllogism假言三段论

- $p \vee q$
  $\neg p$
  $\therefore q$

  Rule of disjunctive syllogism析取三段论



Aristotle
(ca. 384-322 B.C.)

Syllogism Inference Rules

- $p \vee q$
  $\neg p \vee r$
  $\therefore\ q \vee r$

Resolution
消解原理

常见推理定律

- 附加律 $A \Rightarrow (A \vee B)$
- 化简律 $(A \wedge B) \Rightarrow A$
- 假言推理 $(A \rightarrow B) \wedge A \Rightarrow B$
- 拒取式 $(A \rightarrow B) \wedge \neg B \Rightarrow \neg A$
- 析取三段论 $(A \vee B) \wedge \neg B \Rightarrow A$
- 假言三段论 $(A \rightarrow B) \wedge (B \rightarrow C) \Rightarrow (A \rightarrow C)$
- 等价三段论 $(A \leftrightarrow B) \wedge (B \leftrightarrow C) \Rightarrow (A \leftrightarrow C)$
- 构造性两难 $(A \rightarrow B) \wedge (C \rightarrow D) \wedge (A \vee C) \Rightarrow (B \vee D)$
- 构造性两难(特殊形式) $(A \rightarrow B) \wedge (\neg A \rightarrow B) \wedge (A \vee \neg A) \Rightarrow B$
- 破坏性两难 $(A \rightarrow B) \wedge (C \rightarrow D) \wedge (\neg B \vee \neg D) \Rightarrow (\neg A \vee \neg C)$

Formal Proofs

- A formal proof of a conclusion *C*, given premises p_1, p_2,...,p_n consists of a sequence of *steps*, each of which applies some inference rule to premises or previously-proven statements (*antecedents*) to yield a new true statement (the *consequent*).
- A proof demonstrates that *if* the premises are true, *then* the conclusion is true.

推理的形式结构

- 前提： $A_1, A_2, A_3, \ldots, A_k$

  结论： B
- 推理的形式结构：

$$(A_1 \wedge A_2 \wedge A_3 \wedge \ldots \wedge A_k) \rightarrow B$$

- 一个推理是正确的，当且仅当推理的形式结构是永真式

Using Rules of inference to Build Arguments

- The steps of arguments are displayed on separate lines, with the reason for each step explicitly stated.

Formal Proof
Example 6

- Suppose we have the following premises:
  "It is not sunny and it is cold."
  "We will swim only if it is sunny."
  "If we do not swim, then we will canoe."
  "If we canoe, then we will be home early."
- Given these premises, prove the theorem
  "We will be home early" using inference rules.

Proof Example *cont.*

- Let us adopt the following abbreviations:
  - *sunny* = **"It is sunny"**; *cold* = **"It is cold"**; *swim* = **"We will swim"**; *canoe* = **"We will canoe"**; *early* = **"We will be home early"**.
- Then, the premises can be written as:

  (1) $\neg sunny \wedge cold$ (2) $swim \rightarrow sunny$

  (3) $\neg swim \rightarrow canoe$ (4) $canoe \rightarrow early$

Proof Example *cont*.

Step	Reason
1. $\neg sunny \wedge cold$	Premise #1.
2. $\neg sunny$	Simplification of 1.
3. $swim \rightarrow sunny$	Premise #2.
4. $\neg swim$	Modus tollens on 2,3.
5. $\neg swim \rightarrow canoe$	Premise #3.
6. $canoe$	Modus ponens on 4,5.
7. $canoe \rightarrow early$	Premise #4.
8. $early$	Modus ponens on 6,7.

Example 7

- Show that the hypotheses “If you send me an e-mail message, then I will finish writing the program,” “If you do not send me an e-mail message, then I will go to sleep early,” and “If I go to sleep early, then I will wake up feeling refreshed” lead to the conclusion “If I do not finish writing the program, then I will wake up feeling refreshed.”

Proof Example

p: you send me an e-mail message.
q: I will finish writing the program.
r: I will go to sleep early.
s: I will wake up feeling refreshed.

$p \rightarrow q$ $\quad \neg p \rightarrow r$ $\quad r \rightarrow s$ $\quad \neg q \rightarrow s$

Step	Reason
1. $p \rightarrow q$	Premise #1.
2. $\neg q \rightarrow \neg p$	Contrapositive of 1.
3. $\neg p \rightarrow r$	Premise #2.
4. $\neg q \rightarrow r$	Hypothetical syllogism using 2,3.
5. $r \rightarrow s$	Premise #3.
6. $\neg q \rightarrow s$	Hypothetical syllogism using 4,5.

推理（举例）

- 例：下午小王或去看电影或去游泳。他没去看电影。所以，他去游泳了。

  设：　p：小王下午去看电影

  　　q：小王下午去游泳

  前提：$p \vee q$，$\neg p$

  结论：q

证明(举例)

- 前提：$p \vee q$，$\neg p$

  结论：q

  证明：(1) $p \vee q$　　前提引入

  (2) $\neg p$　　前提引入

  (3) q　　(1)(2)析取三段论

推理的形式结构

- 前提： $A_1, A_2, A_3, \ldots, A_k$

  结论： B
- 推理的形式结构：

$$(A_1 \wedge A_2 \wedge A_3 \wedge \ldots \wedge A_k) \rightarrow B$$

- 一个推理是正确的，当且仅当推理的形式结构是永真式

推理定律(deduction laws)

- 推出： $A \Rightarrow B$
  - 读作：A推出B
  - 含义：当A为真时，B也为真
- $A \Rightarrow B$ 当且仅当 $A \rightarrow B$是永真式
- 例如： $(p \vee q) \wedge \neg p \Rightarrow q$

推理定律（举例）

- $(p \vee q) \wedge \neg p \Rightarrow q$
- $(p \vee q) \wedge \neg p \rightarrow q$ 是永真式

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$	$(p \vee q) \wedge \neg p \rightarrow q$
0	0	0	1	0	1
0	1	1	1	1	1
1	0	1	0	0	1
1	1	1	0	0	1

常见推理定律

- 附加律 $A \Rightarrow (A \vee B)$
- 化简律 $(A \wedge B) \Rightarrow A$
- 假言推理 $(A \rightarrow B) \wedge A \Rightarrow B$
- 拒取式 $(A \rightarrow B) \wedge \neg B \Rightarrow \neg A$
- 析取三段论 $(A \vee B) \wedge \neg B \Rightarrow A$
- 假言三段论 $(A \rightarrow B) \wedge (B \rightarrow C) \Rightarrow (A \rightarrow C)$
- 等价三段论 $(A \leftrightarrow B) \wedge (B \leftrightarrow C) \Rightarrow (A \leftrightarrow C)$
- 构造性两难 $(A \rightarrow B) \wedge (C \rightarrow D) \wedge (A \vee C) \Rightarrow (B \vee D)$
- 构造性两难(特殊形式) $(A \rightarrow B) \wedge (\neg A \rightarrow B) \wedge (A \vee \neg A) \Rightarrow B$
- 破坏性两难

$$(A \rightarrow B) \wedge (C \rightarrow D) \wedge (\neg B \vee \neg D) \Rightarrow (\neg A \vee \neg C)$$

推理规则

- 前提引入规则：在证明的任何步骤上都可以引入前提
- 结论引入规则：在证明的任何步骤上所得到的结论都可以做为后继证明的前提
- 置换规则：在证明的任何步骤上，命题公式中的子公式都可以用与之等值的公式置换，得到公式序列中又一个公式
- 分离规则(假言推理) 已知(A→B)，A，则有B。

证明

- 所谓证明，就是一些公式的序列，其中每一个公式或是已知的前提，或是由前面的公式运用推理规则得出的公式，最后一个公式是结论
- 前提： $A_1, A_2, A_3, \ldots, A_k$

  结论： B

  证明：

(1)	A_1	前提引入
(2)	A_2	××××(推理规则)
(3)	A_3	××××
⋮	⋮	⋮
(s)	A_k	××××

证明(举例、续)

- 前提：$(p \wedge q) \rightarrow r$，$\neg s \vee p$，$q$

  结论：$s \rightarrow r$

  证明：

(1)	$\neg s \vee p$	前提引入
(2)	$s \rightarrow p$	(1)置换
(3)	$(p \wedge q) \rightarrow r$	前提引入
(4)	$q \rightarrow (p \rightarrow r)$	(3)置换
(5)	q	前提引入
(6)	$p \rightarrow r$	(4)(5)假言推理
(7)	$s \rightarrow r$	(2)(6)假言三段论

证明(举例、续)

- 证明： $(p \wedge q) \rightarrow r \Leftrightarrow q \rightarrow (p \rightarrow r)$

  $(p \wedge q) \rightarrow r$

  $\Leftrightarrow \neg(p \wedge q) \vee r$ (蕴涵等值式)

  $\Leftrightarrow (\neg p \vee \neg q) \vee r$ (德●摩根律)

  $\Leftrightarrow \neg q \vee (\neg p \vee r)$ (交换律、结合律)

  $\Leftrightarrow q \rightarrow (\neg p \vee r)$ (蕴涵等值式)

  $\Leftrightarrow q \rightarrow (p \rightarrow r)$ (蕴涵等值式)

总结

- 等值式（16组、24条）
  - 幂等律、交换律、结合律、分配律、吸收律；
  - 双重否定律、德●摩根律；
  - 零律、同一律、排中律、矛盾律；
  - 蕴涵等值式、等价等值式、假言易位、等价否定等值式
  - 归谬论
- 推理定律（9条）
  - 附加、化简
  - 假言推理、拒取式、析取三段论、假言三段论、
  - 等价三段论、构造性两难（特殊形式）、破坏性两难

例

- 以下命题已经成立，求谁是作案凶手：
  - (1) 甲或乙作的案
  - (2) 如甲作的案，作案时间应在午夜后
  - (3) 若乙证词正确，则午夜灯光未灭
  - (4) 若乙证词不正确，作案时间不在午夜之后
  - (5) 午夜灯光灭了

例（续）

- 命题符号：
  - A：甲作的案
  - B：乙作的案
  - C：作案时间在午夜后
  - D：乙的证词正确
  - E：午夜灯光灭
  -

- 则以上五句话成为五个命题
  - （1） $A \vee B$
  - （2） $A \rightarrow C$
  - （3） $D \rightarrow \neg E$
  - （4） $\neg D \rightarrow \neg C$
  - （5） E

从而可以演绎出B，即是B作的案。

推理规则

- 附加前提推理规则(简称CP规则)
- 如果 Σ ， $A \Rightarrow B$，则$\Sigma \Rightarrow A \rightarrow B$

- 反证推理规则
- 如果 Σ ， $\neg A \Rightarrow F$，则$\Sigma \Rightarrow A$

例

- 给出¬(A ∧ ¬ B), ¬ B ∨ C, ¬ C ⇒ ¬ A的证明

例

- 给出的证明$A_1 \rightarrow (A_2 \rightarrow A_3), A_4 \rightarrow A_1, A_2 \Rightarrow A_4 \rightarrow A_3$

例

- 给出的证明$A_1 \to \neg A_2$, $A_2 \vee \neg A_3$,
- $A_3 \wedge \neg A_4 \Rightarrow \neg A_1$

习题

- 证明下面的等值式：

  (1) $(p \rightarrow q) \wedge (p \rightarrow r) \Leftrightarrow p \rightarrow (q \wedge r)$

  (2) $(p \wedge \neg q) \vee (\neg p \wedge q) \Leftrightarrow (p \vee q) \wedge \neg (p \wedge q)$

- 构造下面推理的证明：

  (1) 前提：$p \rightarrow (q \rightarrow r)$，$p$，$q$

  结论：$r \vee s$

  (2) 前提：$p \rightarrow q$，$\neg (q \wedge r)$，r

  结论：$\neg p$

例：

- *有一个逻辑学家误入某部落，被拘于劳狱，酋长意欲放行，他对逻辑学家说：“有两门，一为自由，一为死亡，你可任意开启一门。为协助你脱逃，今加派两名战士负责解答你所提的一个问题。惟可虑者，此两战士中一名天性诚实，一名说谎成性，今后生死由你自己选择。”逻辑学家沉思片刻，即向一战士发问，然后开门从容离去。该逻辑学家应如何发问？*

解：

- 逻辑学家手指一门问身旁的一名战士说：“这扇门是死亡门，他（指另一名战士）将答‘是’，对吗？”
- 当被问战士回答“对”，则逻辑学家开启所指的门从容离去。当回答“否”，则开启另一扇门从容离去。

分析

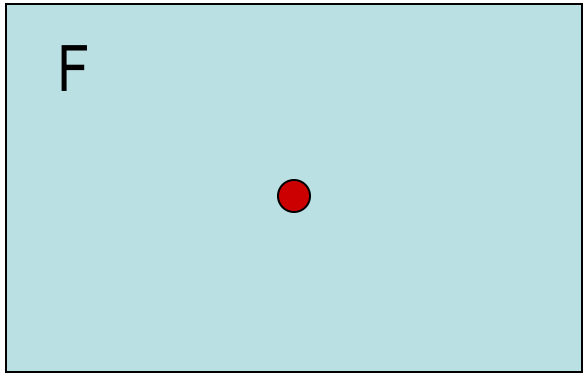
- P：被问战士是诚实人
- Q：被问战士的回答是“是”
- R：另一名战士的回答是“是”
- S：这扇门是死亡之门

P	Q	R	S
0	0	1	1
0	1	0	0
1	0	0	1
1	1	1	0

一阶谓词推理证明

一阶逻辑推理定律(定义)

- 推出： $A \Rightarrow B$
  - 读作：A推出B
  - 含义：A为真时, B也为真
- $A \Rightarrow B$ 当且仅当 $A \rightarrow B$是永真式
- 例如： $\forall x F(x) \Rightarrow \exists x F(x)$



一阶逻辑推理定律(来源)

- 命题逻辑推理定律的代换实例
- 基本等值式生成的推理定律
- 其他的一阶逻辑推理定律

$\forall x A(x) \vee \forall x B(x) \Rightarrow \forall x (A(x) \vee B(x))$

$\exists x (A(x) \wedge B(x)) \Rightarrow \exists x A(x) \wedge \exists x B(x)$

$\forall x (A(x) \rightarrow B(x)) \Rightarrow \forall x A(x) \rightarrow \forall x B(x)$

$\exists x (A(x) \rightarrow B(x)) \Rightarrow \exists x A(x) \rightarrow \exists x B(x)$

$\vdots$

一阶逻辑推理定律(举例)

- 命题逻辑推理定律的代换实例

  例如: 假言推理规则:

  $$(A \rightarrow B) \wedge A \Rightarrow B$$

  代入 A=F(a), B=G(a), 得到

  $$(F(a) \rightarrow G(a)) \wedge F(a) \Rightarrow G(a)$$

一阶逻辑推理定律(举例、续)

- 基本等值式生成的推理定律

  即由 $A \Leftrightarrow B$ 可得 $A \Rightarrow B$ 和 $B \Rightarrow A$

  例如: 量词分配等值式:

  $\forall x(A(x) \wedge B(x)) \Leftrightarrow \forall xA(x) \wedge \forall xB(x)$

  可得

  $\forall x(A(x) \wedge B(x)) \Rightarrow \forall xA(x) \wedge \forall xB(x)$

  $\forall xA(x) \wedge \forall xB(x) \Rightarrow \forall x(A(x) \wedge B(x))$

推理定律与推理规则

- 推理定律$A \Rightarrow B$表示$A \rightarrow B$是永真式
- 推理规则是在证明过程中使用的规则
- 每一条推理定律都可以作为推理规则
- 有些推理规则不是推理定律

一阶逻辑的常用推理规则

- 前提引入、结论引入、置换规则
- 假言推理、附加、化简、拒取式、假言三段论、析取三段论、构造性两难、合取引入
- *UI*、*UG*、*EI*、*EG*

Inference Rules for Quantifiers

- $\forall x\, P(x)$
  $\therefore P(o)$ (substitute *any* specific object *o*)
  Universal instantiation
- $P(g)$ (for *g* a *general* element of u.d.)
  $\therefore \forall x\, P(x)$
  Universal generalization
- $\exists x\, P(x)$
  $\therefore P(c)$ (substitute a *new constant c*)
  Existential instantiation
- $P(o)$ (substitute any extant object *o*)
  $\therefore \exists x\, P(x)$
  Existential generalization

UI规则(universal instantiation)全称量词消去规则

- 表示为　　$\forall xA(x)$　　$\forall xA(x)$

　　——————　或　——————

　　$\therefore A(y)$　　$\therefore A(c)$

- 注意1: y是自由变项； c是个体常项
- 注意2: 被消去量词的辖域是整个公式
- 例如

(1) $\forall x(F(x)\rightarrow G(x))$　　前提引入

(2) $F(a)\rightarrow G(a)$　　(1)UI

UG规则(universal generalization)
全称量词引入规则

- 表示为 $\dfrac{A(y)}{\therefore \forall x A(x)}$
- 注意1: y是自由变项
- 注意2: 量词加在整个公式前面
- 例如

(1) $F(y)\rightarrow G(y)$ 前提引入

(2) $\forall x(F(x)\rightarrow G(x))$ (1)UG

EI规则(existential instantiation)存在量词消去规则

- 表示为 $\exists x A(x)$

$$\therefore A(c)$$

- 注意1: c是特定的满足A的个体常项
- 注意2: 被消去量词的辖域是整个公式
- 例如

(1) $\exists x(F(x) \wedge G(x))$ 前提引入

(2) $F(a) \wedge G(a)$ (1)EI

EG规则(existential generalization)存在量词引入规则

- 表示为

$$\frac{A(c)}{\therefore \exists x A(x)}$$

- 注意1：c是个体常项
- 注意2：量词加在整个公式前面
- 例如

(1) $F(c) \wedge G(c)$ 前提引入

(2) $\exists x(F(x) \wedge G(x))$ (1)EG

构造推理的证明(举例)

- 前提: $\forall x(F(x)\rightarrow G(x))$，$F(a)$

  结论: $G(a)$

  证明: (1) $F(a)$ 前提引入

  (2) $\forall x(F(x)\rightarrow G(x))$ 前提引入

  (3) $F(a)\rightarrow G(a)$ (2)UI

  (4) $G(a)$ (1)(3)假言推理

构造推理的证明(举例、续)

- 前提: $\forall x(F(x)\rightarrow G(x))$，$\exists xF(x)$

  结论: $\exists xG(x)$

  证明:

(1)	$\exists xF(x)$	前提引入
(2)	$F(c)$	(1) EI
(3)	$\forall x(F(x)\rightarrow G(x))$	前提引入
(4)	$F(c)\rightarrow G(c)$	(3) UI
(5)	$G(c)$	(2)(4)假言推理
(6)	$\exists xG(x)$	(5) EG

构造推理的证明(举例、续)

- “先EI，后UI”
- 证明: (1) $\forall x(F(x)\rightarrow G(x))$ 前提引入
  (2) $F(c)\rightarrow G(c)$ (2) UI
  (3) $\exists xF(x)$ 前提引入
  (4) $F(c)$ (3) EI
  ⋮ ⋮ ⋮
- 说明：这个证明是错的. (3)(4)应当在(1)(2)前，(4)中的c是特定的， (2)中的c是任意的

Example 13

- Show that the premises “A student in this class has not read the book,” and “ Everyone in this class passed the first exam” imply the conclusion “Someone who passed the first exam has not read the book.”

Example 13

Premises

- A student in this class has not read the book
- $\exists x(C(x) \wedge \neg B(x))$
- Everyone in this class passed the exam $\forall x(C(x) \rightarrow P(x))$

Conclusion

- Someone who passed the exam has not read the book. $\exists x(P(x) \wedge \neg B(x))$
- C(x):x in this class B(x):x read the book
- P(x):x pass the exam

Example 13

step	Reason
1 $\exists x(C(x) \wedge \neg B(x))$	Premise
2 $C(a) \wedge \neg B(a)$	EI
3 $C(a)$	Simplification from (2)
4 $\forall x(C(x) \rightarrow P(x))$	Premise
5 $C(a) \rightarrow P(a)$	UI
6 $P(a)$	Modus ponens from(3)and(5)
7 $\neg B(a)$	Simplification from (2)
8 $P(a) \wedge \neg B(a)$	Conjunction from(6) and (7)
9 $\exists x(P(x) \wedge \neg B(x))$	EG

构造推理的证明(举例、续)

- 前提: $\neg\exists x(F(x)\wedge H(x))$, $\forall x(G(x)\rightarrow H(x))$,
  结论: $\forall x(G(x)\rightarrow\neg F(x))$
- 证明: (1) $\neg\exists x(F(x)\wedge H(x))$ 前提引入
  (2) $\forall x(\neg F(x)\vee\neg H(x))$ (1)置换
  (3) $\forall x(H(x)\rightarrow\neg F(x))$ (2)置换
  (4) $H(y)\rightarrow\neg F(y)$ (3) UI
  (5) $\forall x(G(x)\rightarrow H(x))$ 前提引入
  (6) $G(y)\rightarrow H(y)$ (5) UI
  (7) $G(y)\rightarrow\neg F(y)$ (4)(6)假言三段论
  (8) $\forall x(G(x)\rightarrow\neg F(x))$ (7) UG

Exercise

- Show that the premises “ Everyone who likes walking doesn‘t like riding a bicycle. ” ; “ Everyone likes riding a bicycle or riding a car. ” ; “ Some people don’t like riding a car.” imply the conclusion “ Some people don‘t like walking.”
- (The **universe of discourse** is a collection of all human beings)

习题 - 构造下面推理的证明

- (1) 前提：$\exists xF(x)\rightarrow\forall y((F(y)\vee G(y))\rightarrow R(y))$,
  $\exists xF(x)$
  结论： $\exists xR(x)$
- (2) 前提：$\forall x(F(x)\vee G(x))$,
  $\forall x(\neg G(x)\vee\neg R(x))$,
  $\forall xR(x)$
  结论： $\forall xF(x)$

(3)有些病人相信所有的医生，但病人都不相信一个骗子。证明：医生都不是骗子。

§ 1.6 – 4, 6, 12, 16, 20, 24

1.7 Introduction to Proofs
定理证明方法：

- **1.7.1 Introduction**
- **1.7.2 Some Terminology**
- **1.7.3 Understanding How Theorems Are Stated**
- **1.7.4 Methods of Proving Theorems**
- **1.7.5 Direct Proofs**
- **1.7.6 Proof by Contraposition**
- **1.7.7 Proofs by Contradiction**
- **1.7.8 Mistakes in Proofs**

Applications of Proofs

- An exercise in clear communication of logical arguments in any area of study.
- The fundamental activity of mathematics is the discovery and elucidation, through proofs, of interesting new theorems.
- Theorem-proving has applications in program verification, computer security, automated reasoning systems, *etc.*
- Proving a theorem allows us to rely upon on its correctness even in the most critical scenarios.

Proof Terminology术语

- *Theorem定理*
  - A statement that has been proven to be true.
- *Axioms*, postulates, *hypotheses*, *premises公理 假定 假设 前提*
  - Assumptions (often unproven) defining the structures about which we are reasoning.
- *Rules of inference 推理规则*
  - Patterns of logically valid deductions from hypotheses to conclusions.

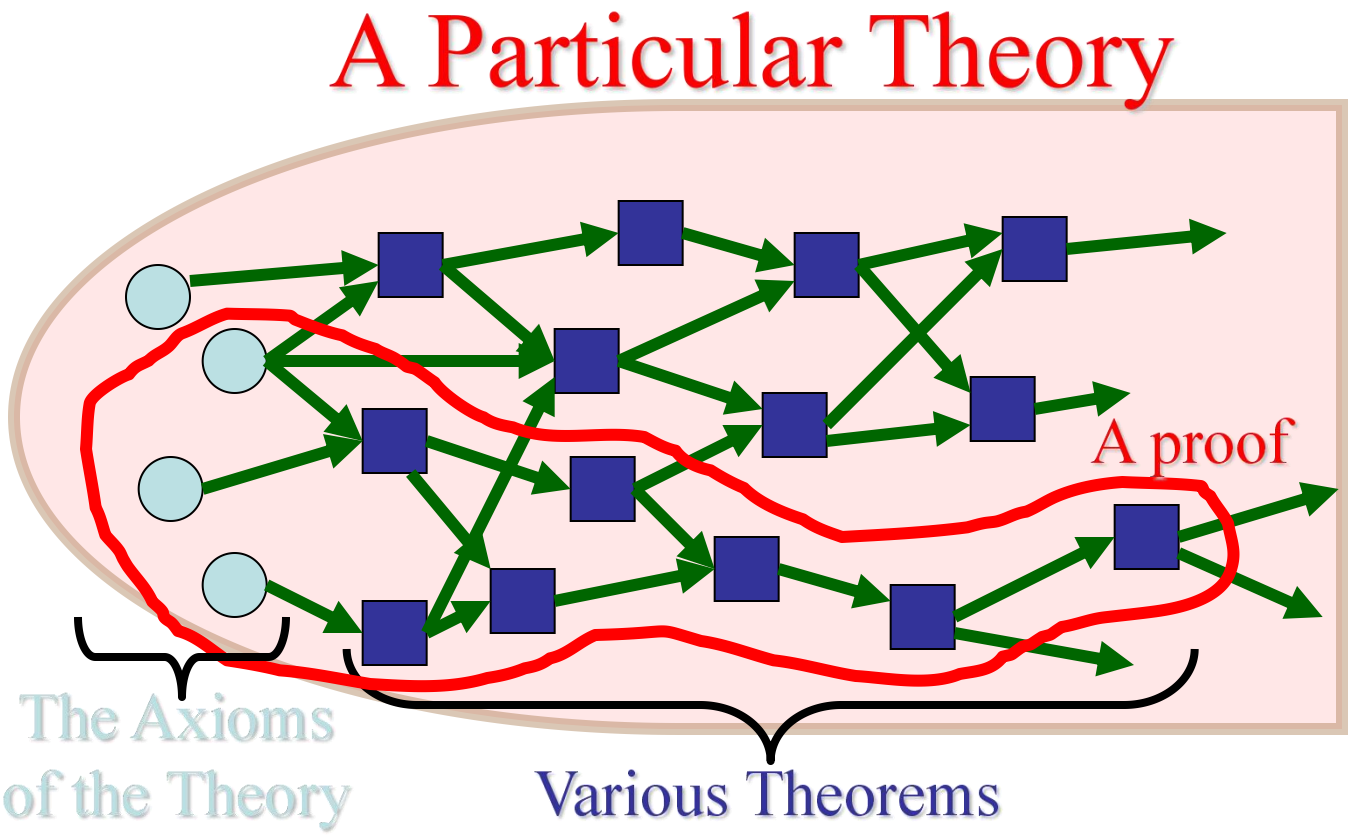
More Proof Terminology

- *Lemma* 引理- A minor theorem used as a stepping-stone to proving a major theorem.
- *Corollary* 推论- A minor theorem proved as an easy consequence of a major theorem.
- *Conjecture* 猜想- A statement whose truth value has not been proven. (A conjecture may be widely believed to be true, regardless.)
- *Theory* 理论– The set of all theorems that can be proven from a given set of axioms.

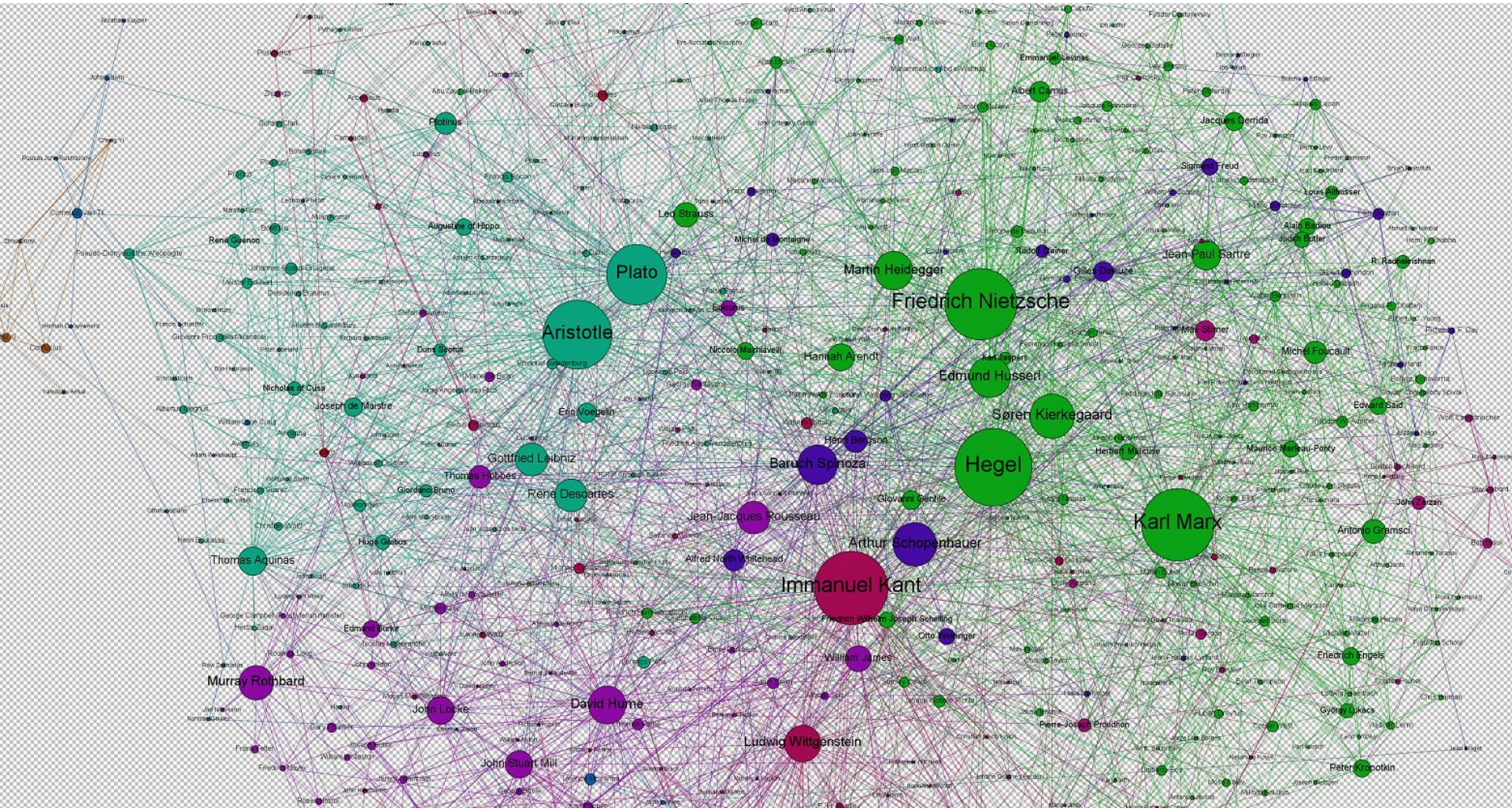
推理（deduction）

- 推理：从前提出发推出结论的思维过程
- 前提：已知命题公式的集合
- 结论：从前提出发应用推理规则推出的命题公式

Graphical Visualization



Axioms
Outlook
Theorem
Thought
Doctrine
Ideology


Plato
Aristotle
Friedrich Nietzsche
Hegel
Immanuel Kant
Karl Marx
Martin Heidegger
Edmund Husserl
Baruch Spinoza
Jean-Jacques Rousseau
Arthur Schopenhauer
René Descartes
Gottfried Leibniz
Thomas Hobbes
David Hume
John Locke
Ludwig Wittgenstein
John Stuart Mill
Thomas Aquinas
Augustine of Hippo
Plotinus
Leo Strauss
Hannah Arendt
Søren Kierkegaard
Jean-Paul Sartre
Michel Foucault
Gilles Deleuze
Jacques Derrida
Emmanuel Levinas
Albert Camus
Sigmund Freud
Louis Althusser
Alain Badiou
Judith Butler
Slavoj Žižek
Eric Voegelin
Joseph de Maistre
Nicholas of Cusa
Niccolò Machiavelli
Michel de Montaigne
Desiderius Erasmus
Duns Scotus
René Guénon
Henri Bergson
Herbert Marcuse
Maurice Merleau-Ponty
Edward Said
Antonio Gramsci
Friedrich Engels
György Lukács
Peter Kropotkin
Pierre-Joseph Proudhon
Otto Weininger
William James
Friedrich Wilhelm Joseph Schelling
Alfred North Whitehead
Giovanni Gentile
Hugo Grotius
Edmund Burke
Murray Rothbard
Francisco Suárez
Giordano Bruno
Thomas Malthus
Karl Jaspers
Max Stirner
R. Radhakrishnan
John Zerzan

1.7 Introduction to Proofs
定理证明方法：

- 1、直接证明/direct proof:　　$p \rightarrow Q$
- 2、间接证明/indirect proof : $p \rightarrow Q \Leftrightarrow \neg Q \rightarrow \neg P$
- 3、空证明/vacuous proof:　　$p \rightarrow Q$ 其中 P为 F
- 4、平凡证明/trivial proof：　　$p \rightarrow Q$ 其中 Q 为T

定理证明方法：

- 5、反证明/proof of contradiction：
- P
- $\neg P \Rightarrow S \wedge \neg S$
- 6、分例证明/proof of cases：
- $P_1 \vee P_2 \ldots \vee P_n \rightarrow Q$
- $(P_1 \rightarrow Q) \wedge (P_2 \rightarrow Q) \ldots \wedge (P_n \rightarrow Q)$

定理证明方法：（含有量词）

- 7、存在证明/existence proof：
- $\exists x\, P(x)$
- 8、归纳证明/induction proof ：
- $\forall x\, P(x)$

Proof Methods for Implications

For proving implications $p \rightarrow q$, we have:

- *Direct* proof: Assume p is true, and prove q.
- *Indirect* proof: Assume $\neg q$, and prove $\neg p$.
- *Vacuous* proof: Prove $\neg p$ by itself.
- *Trivial* proof: Prove q by itself.
- Proof by cases:
  Show $p \rightarrow (a \vee b)$, and $(a \rightarrow q)$ and $(b \rightarrow q)$.

Direct Proof Example

- **Definition:** An integer n is called *odd* iff $n=2k+1$ for some integer k; n is *even* iff $n=2k$ for some k.
- **Theorem:** Every integer is either odd or even.
  - This can be proven from even simpler axioms.
- **Theorem:** (For all numbers n) If n is an odd integer, then n^2 is an odd integer.
- **Proof:** If n is odd, then $n = 2k+1$ for some integer k. Thus, $n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$. Therefore n^2 is of the form $2j + 1$ (with j the integer $2k^2 + 2k$), thus n^2 is odd. □

Indirect Proof Example

- **Theorem:** (For all integers n)
  If $3n+2$ is odd, then n is odd.
- **Proof:** Suppose that the conclusion is false, *i.e.*, that n is even. Then $n=2k$ for some integer k. Then $3n+2 = 3(2k)+2 = 6k+2 = 2(3k+1)$. Thus $3n+2$ is even, because it equals $2j$ for integer $j = 3k+1$. So $3n+2$ is not odd. We have shown that $\neg(n \text{ is odd}) \rightarrow \neg(3n+2 \text{ is odd})$, thus its contra-positive $(3n+2 \text{ is odd}) \rightarrow (n \text{ is odd})$ is also true. □

Vacuous Proof Example

- **Theorem:** (For all n) If n is both odd and even, then $n^2 = n + n$.
- **Proof:** The statement "n is both odd and even" is necessarily false, since no number can be both odd and even. So, the theorem is vacuously true. □

Trivial Proof Example

- **Theorem:** (For integers *n*) If *n* is the sum of two prime numbers, then either *n* is odd or *n* is even.
- **Proof:** *Any* integer *n* is either odd or even. So the conclusion of the implication is true regardless of the truth of the antecedent. Thus the implication is true trivially. □

Proof by Contradiction

- A method for proving p.
- Assume $\neg p$, and prove both q and $\neg q$ for some proposition q. (Can be anything!)
- Thus $\neg p \rightarrow (q \wedge \neg q)$
- $(q \wedge \neg q)$ is a trivial contradiction, equal to **F**
- Thus $\neg p \rightarrow \mathbf{F}$, which is only true if $\neg p = \mathbf{F}$
- Thus p is true.

Proof by Contradiction Example

- **Theorem:** $\sqrt{2}$ is irrational.
  - **Proof:** Assume $2^{1/2}$ were rational. This means there are integers i,j with no common divisors such that $2^{1/2} = i/j$. Squaring both sides, $2 = i^2/j^2$, so $2j^2 = i^2$. So i^2 is even; thus i is even. Let $i=2k$. So $2j^2 = (2k)^2 = 4k^2$. Dividing both sides by 2, $j^2 = 2k^2$. Thus j^2 is even, so j is even. But then i and j have a common divisor, namely 2, so we have a contradiction. □

$\sqrt{2}$

Review: Proof Methods So Far

- *Direct*, *indirect*, *vacuous*, and *trivial* proofs of statements of the form $p \rightarrow q$.
- *Proof by contradiction* of any statements.
- Next: *Constructive* and *nonconstructive existence proofs*.

Proving Existentials

- A proof of a statement of the form $\exists x\, P(x)$ is called an *existence proof*.
- If the proof demonstrates how to actually find or construct a specific element *a* such that $P(a)$ is true, then it is a *constructive* proof.
- Otherwise, it is *nonconstructive*.

Constructive Existence Proof

- **Theorem:** There exists a positive integer n that is the sum of two perfect cubes in two different ways:
  - equal to $j^3 + k^3$ and $l^3 + m^3$ where j, k, l, m are positive integers, and $\{j,k\} \neq \{l,m\}$
- **Proof:** Consider $n = 1729$, $j = 9$, $k = 10$, $l = 1$, $m = 12$. Now just check that the equalities hold.

Another Constructive Existence Proof

- **Theorem:** For any integer $n>0$, there exists a sequence of n consecutive composite integers.
- Same statement in predicate logic:
  $\forall n>0\ \exists x\ \forall i\ (1\leq i\leq n)\rightarrow(x+i \text{ is composite})$
- Proof follows on next slide…

The proof...

- Given $n>0$, let $x = (n + 1)! + 1$.
- Let $i \geq 1$ and $i \leq n$, and consider $x+i$.
- Note $x+i = (n + 1)! + (i + 1)$.
- Note $(i+1)|(n+1)!$, since $2 \leq i+1 \leq n+1$.
- Also $(i+1)|(i+1)$. So, $(i+1)|(x+i)$.
- $\therefore$ $x+i$ is composite.
- $\therefore$ $\forall n \, \exists x \, \forall 1 \leq i \leq n$: $x+i$ is composite. Q.E.D.

Nonconstructive Existence Proof

- **Theorem:**
  "There are infinitely many prime numbers."
- Any finite set of numbers must contain a maximal element, so we can prove the theorem if we can just show that there is *no* largest prime number.
- *I.e.*, show that for any prime number, there is a larger number that is *also* prime.
- More generally: For *any* number, ∃ a larger prime.
- Formally: Show $\forall n \; \exists p > n$: p is prime.

The proof, using *proof by cases*...

- Given $n>0$, prove there is a prime $p>n$.
- Consider $x = n!+1$. Since $x>1$, we know (x is prime)$\vee$(x is composite).
- **Case 1:** x is prime. Obviously $x>n$, so let $p=x$ and we're done.
- **Case 2:** x has a prime factor p. But if $p\leq n$, then p mod $x = 1$. So $p>n$, and we're done.

Limits on Proofs

- Some very simple statements of number theory haven't been proved or disproved!
  - *E.g. Goldbach's conjecture*: Every integer $n \geq 2$ is exactly the average of some two primes.
  - $\forall n \geq 2 \; \exists$ primes p,q: $n=(p+q)/2$.
- There are true statements of number theory (or any sufficiently powerful system) that can *never* be proved (or disproved) (Gödel).

Common Fallacies

- A *fallacy* is an inference rule or other proof method that is not logically valid.
  - A fallacy may yield a false conclusion!
- Fallacy of *affirming the conclusion*:
  - "$p \rightarrow q$ is true, and q is true, so p must be true." (No, because $\mathbf{F} \rightarrow \mathbf{T}$ is true.)
- Fallacy of *denying the hypothesis*:
  - "$p \rightarrow q$ is true, and p is false, so q must be false." (No, again because $\mathbf{F} \rightarrow \mathbf{T}$ is true.)

Circular Reasoning

- The fallacy of (explicitly or implicitly) assuming the very statement you are trying to prove in the course of its proof. Example:
- Prove that an integer n is even, if n^2 is even.
- **Attempted proof:** "Assume n^2 is even. Then $n^2=2k$ for some integer k. Dividing both sides by n gives $n = (2k)/n = 2(k/n)$. So there is an integer j (n
  Therefore n is e
  - Circular reasoni

*Begs the question: How do you show that $j=k/n=n/2$ is an integer, without **first** assuming that n is even?*

Proof Methods and Strategy

Section 1.8

Proof Methods and Strategy

Rosen 8th ed., § 1.8

1.8 Proof Methods and Strategy

- 1.8.1 Introduction
- 1.8.2 Exhaustive Proof and Proof by Cases
- 1.8.3 Existence Proofs
- 1.8.4 Uniqueness Proofs
- 1.8.5 Proof Strategies
- 1.8.6 Looking for Counterexamples
- 1.8.7 Proof Strategy in Action
- 1.8.8 Tilings
- 1.8.9 The Role of Open Problems
- 1.8.10 Additional Proof Methods

Overview

- In 1.7, we already saw:
  - Several types of proofs of implications $p \rightarrow q$:
    - Vacuous, Trivial, Direct, Indirect
  - Types of existence proofs:
    - Constructive vs. Nonconstructive.
  - Some methods of proving general statements p:
    - Proof by cases, proof by contradiction.
- In this module, we will see examples of:
  - Forward vs. backward reasoning.
  - Proof by cases.
  - Adapting existing proofs.
  - Turning conjectures into proofs.

Forward Reasoning

- Have premises p, and want to prove q.
  - Find a s_1 such that $p \rightarrow s_1$
    - Then, modus ponens gives you s_1.
  - Then, find an s_2 ∋ (such that) $s_1 \rightarrow s_2$.
    - Then, modus ponens gives you s_2.
  - And hope to eventually get to an s_n ∋ $s_n \rightarrow q$.
- The problem with this method is…
  - It can be tough to see the path looking from p.

Backward Reasoning

- It can often be easier to see the *very same path* if you just start looking from the conclusion *q* instead…
  - That is, *first* find an s_{-1} such that $s_{-1} \rightarrow q$.
  - *Then*, find an $s_{-2} \ni s_{-2} \rightarrow s_{-1}$, and so on…
  - Working back to an $s_{-n} \ni p \rightarrow s_{-n}$.
- Note we *still* are using *modus ponens* to propagate truth *forwards* down the chain from *p* to s_{-n} to … to s_1 to *q*!
  - We are *finding* the chain *backwards*, but *applying* it *forwards*.
  - This is not quite the same thing as an indirect proof…
    - In that, we would use *modus tollens* and $\neg q$ to prove $\neg s_{-1}$, *etc.*
  - However, it it similar.

Backward Reasoning Example

Example 1

- **Theorem:**

  $\forall a>0, b>0, a\neq b: (a+b)/2 > (ab)^{1/2}$.

- **Proof:**
  - Notice it is not obvious how to go from the premises $a>0$, $b>0$, $a\neq b$ directly forward to the conclusion $(a+b)/2 > (ab)^{1/2}$.
  - So, let's work *backwards* from the conclusion, $(a+b)/2 > (ab)^{1/2}$!

Steps of Example

- $(a+b)/2 > (ab)^{1/2} \Leftrightarrow$ (squaring both sides)
  - This preserves the "$>$" since both sides are positive.
- $(a+b)^2/4 > ab \Leftrightarrow$ (multiplying through by 4)
- $(a+b)^2 > 4ab \Leftrightarrow$ (squaring $a+b$)
- $a^2+2ab+b^2 > 4ab \Leftrightarrow$ (subtracting out $4ab$)
- $a^2-2ab+b^2 > 0 \Leftrightarrow$ (factoring left side)
- $(a-b)^2 > 0$
- Now, since $a \neq b$, $(a-b) \neq 0$, thus $(a-b)^2 > 0$, and we can work our way back along the chain of steps…

"Forwardized" version of Example

- **Theorem:** $\forall a>0, b>0, a\neq b: (a+b)/2 > (ab)^{1/2}$.
  - **Proof.** If Since $a\neq b$, $(a-b)\neq 0$. Thus, $(a-b)^2>0$, *i.e.*, $a^2-2ab+b^2 > 0$. Adding $4ab$ to both sides, $a^2+2ab+b^2 > 4ab$. Factoring the left side, we have $(a+b)^2 > 4ab$, so $(a+b)^2/4 > ab$. Since ab is positive, we can take the square root of both sides and get $(a+b)/2 > (ab)^{1/2}$. ■
- This is just a simple proof proceeding directly from premises to conclusion, but if you don't realize how it was obtained, it looks "magical."
  - A common student reaction: "But how did you *know* to pick $4ab$ out of thin air, to add to both sides?"
    - Answer: By working backwards from the conclusion!

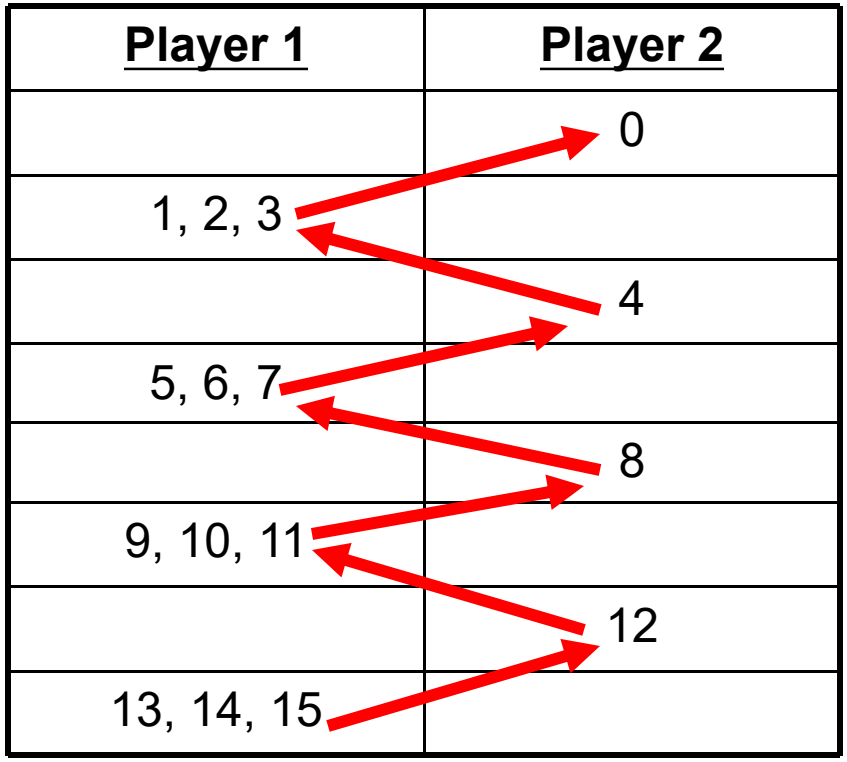
Stone Game Example

Example 2

- Game rules:
  - There are 15 stones in a pile. Two players take turns removing either 1, 2, or 3 stones. Whoever takes the last stone wins.
- **Theorem:** There is a strategy for the first player that guarantees him a win.
- How do we prove this? Constructive proof…
  - Looks complicated… How do we pick out the winning strategy from among all possible strategies?
    - Work backwards from the endgame!

Working Backwards in the Game

- Player 1 wins if it is player 2's turn and there are no stones…
- P1 can arrange this if it is his turn, and there are 1, 2, or 3 stones…
- This will be true as long as player 2 had 4 stones on his turn…
- And so on…



“Forwardized” version

- **Theorem.** Whoever moves first can always force a win.
  - **Proof.** Player 1 can remove 3 stones, leaving 12. After player 2 moves, there will then be either 11, 10, or 9 stones left. In any of these cases, player 1 can then reduce the number of stones to 8. Then, player 2 will reduce the number to 7, 6, or 5. Then, player 1 can reduce the number to 4. Then, player 2 must reduce them to 3, 2, or 1. Player 1 then removes the remaining stones and wins.

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Proof by Cases

- To prove a conditional statement of the form:

$$(p_1 \vee p_2 \vee \ldots \vee p_n) \rightarrow q$$

- Use the tautology

$$[(p_1 \vee p_2 \vee \ldots \vee p_n) \rightarrow q] \leftrightarrow$$
$$[(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \ldots \wedge (p_n \rightarrow q)]$$

- Each of the implications $p_i \rightarrow q$ is a *case*.

当前无法显示此图像。

Proof by Cases

Example: Let $a \,@\, b = \max\{a, b\} = a$ if $a \geq b$, otherwise $a \,@\, b = \max\{a, b\} = b$.

Show that for all real numbers a, b, c

$$(a \,@\, b) \,@\, c = a \,@\, (b \,@\, c)$$

(This means the operation @ is associative.)

Proof: Let a, b, and c be arbitrary real numbers.

Then one of the following 6 cases must hold.

1. $a \geq b \geq c$
2. $a \geq c \geq b$
3. $b \geq a \geq c$
4. $b \geq c \geq a$
5. $c \geq a \geq b$
6. $c \geq b \geq a$

Continued on next slide →

Proof by Cases

Case 1: $a \geq b \geq c$

$(a \mathbin{@} b) = a$, $a \mathbin{@} c = a$, $b \mathbin{@} c = b$

Hence $(a \mathbin{@} b) \mathbin{@} c = a = a \mathbin{@} (b \mathbin{@} c)$

Therefore the equality holds for the first case.

A complete proof requires that the equality be shown to hold for all 6 cases. But the proofs of the remaining cases are similar. Try them.

Without Loss of Generality

Example: Show that if x and y are integers and both $x \cdot y$ *and* $x+y$ are even, then both x and y are even.

Proof: Use a proof by contraposition. Suppose x and y are not both even. Then, one or both are odd. Without loss of generality, assume that x is odd. Then $x = 2m + 1$ for some integer m.

Case 1: y is even. Then $y = 2n$ for some integer n, so $x + y = (2m + 1) + 2n = 2(m + n) + 1$ is odd.

Case 2: y is odd. Then $y = 2n + 1$ for some integer n, so $x \cdot y = (2m + 1)(2n + 1) = 2(2m \cdot n + m + n) + 1$ is odd.

◀

We only cover the case where x is odd because the case where y is odd is similar. The use phrase *without loss of generality* (WLOG) indicates this.

Proof by Cases Example

Example 3

- **Theorem:** $\forall n \in \mathbf{Z}\ \neg(2|n \vee 3|n) \rightarrow 24|(n^2-1)$
  - **Proof:** Since $2\cdot3=6$, the value of n mod 6 is sufficient to tell us whether $2|n$ or $3|n$. If (n mod 6)$\in\{0,3\}$ then $3|n$; if it is in $\{0,2,4\}$ then $2|n$. Thus (n mod 6)$\in\{1,5\}$.
    - **Case #1:** If n mod 6 = 1, then $(\exists k)\ n=6k+1$. $n^2=36k^2+12k+1$, so $n^2-1=36k^2+12k = 12(3k+1)k$. Note $2|(3k+1)k$ since either k or $3k+1$ is even. Thus $24|(n^2-1)$.
    - **Case #2:** If n mod 6 = 5, then $n=6k+5$. $n^2-1 = (n-1)\cdot(n+1) = (6k+4)\cdot(6k+6) = 12\cdot(3k+2)\cdot(k+1)$. Either $k+1$ or $3k+2$ is even. Thus, $24|(n^2-1)$.

Proof by Examples?

- A universal statement can never be proven by using examples, unless the universe can be validly reduced to only finitely many examples, and your proof covers all of them!
- **Theorem:** $\neg\exists x,y\in\mathbf{Z}: x^2+3y^2 = 8$.
  - **Proof:** If $|x|\geq 3$ or $|y|\geq 2$ then $x^2+3y^2 > 8$. This leaves $x^2\in\{0,1,4\}$ and $3y^2\in\{0,3\}$. The largest pair sum to $4+3 = 7 < 8$.

Example 4

EXAMPLE 5 Formulate a conjecture about the final decimal digit of the square of an integer and prove your result.

- Solution: The smallest perfect squares are 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169,196, 225, and so on. We notice that the digits that occur as the final digit of a square are 0, 1, 4, 5, 6, and 9, with 2, 3, 7, and 8 never appearing as the final digit of a square. We conjecturethis theorem: The final decimal digit of a perfect square is 0, 1, 4, 5, 6, or 9. How can we prove this theorem?

- We first note that we can express an integer n as $10a + b$, where a and b are positive integers and b is 0, 1, 2, 3, 4, 5, 6, 7, 8, or 9. Here a is the integer obtained by subtracting the final decimal digit of n from n and dividing by 10. Next, note that $(10a + b)^2 = 100a^2 + 20ab + b^2 =$
- $10(10a^2 + 2b) + b^2$, so that the final decimal digit of n^2 is the same as the final decimal digit of b^2. Furthermore, note that the final decimal digit of b^2 is the same as the final decimal digit of $(10 - b)^2 = 100 - 20b + b^2$. Consequently, we can reduce our proof to the consideration of six cases

A Constructive Existence Proof

Example 7

- **Theorem:** For any integer $n>0$, there exists a sequence of n consecutive composite integers.
- Same statement in predicate logic:
  $\forall n>0\ \exists x\ \forall i\ (1\le i\le n)\rightarrow(x+i \text{ is composite})$
- Proof follows on next slide…

The proof...

- Given $n>0$, let $x = (n + 1)! + 1$.
- Let $i \geq 1$ and $i \leq n$, and consider $x+i$.
- Note $x+i = (n + 1)! + (i + 1)$.
- Note $(i+1)|(n+1)!$, since $2 \leq i+1 \leq n+1$.
- Also $(i+1)|(i+1)$. So, $(i+1)|(x+i)$.
- $\therefore$ $x+i$ is composite.
- $\therefore$ $\forall n \, \exists x \, \forall 1 \leq i \leq n : x+i$ is composite. Q.E.D.

Nonconstructive Existence Proof

- **Theorem:** There are infinitely many prime numbers.
  - Any finite set of numbers must contain a maximal element, so we can prove the theorem if we can just show that there is no *largest* prime number.
  - *I.e.*, show that for any prime number, there is a larger number that is *also* prime.
  - More generally: For *any* number, ∃ a larger prime.
  - Formally: Show $\forall n \; \exists p > n : p$ is prime.

The proof, using *proof by cases*…

- Given $n>0$, prove there is a prime $p>n$.
- Consider $x = n!+1$. Since $x>1$, we know that (x is prime)$\vee$(x is composite).
- **Case 1:** Suppose x is prime. Obviously $x>n$, so let $p=x$ and we're done.
- **Case 2:** x has a prime factor p. But if $p\leq n$, then $p \bmod x = 1$. So $p>n$, and we're done.

Adapting Existing Proofs

Example 5

- **Theorem:** There are infinitely many primes of the form $4k+3$, where $k \in \mathbf{N}$.
  - Recall we proved there are infinitely many primes because if $p_1,\ldots,p_n$ were all the primes, then $(\prod p_i)+1$ must be prime or have a prime factor greater than p_n, $\Rightarrow$ contradiction!
  - **Proof:** Similarly, suppose $q_1,\ldots,q_n$ lists all primes of the form $4k+3$,
    - and analogously consider $Q = 4(\prod q_i)+3$.
      - Unfortunately, since $q_1 = 3$ is possible, $3|Q$ and so Q does have a prime factor among the q_i, so this doesn't work!
    - So instead, consider $Q = 4(\prod q_i)-1 = 4(\prod q_i-1)+3$. This has the right form, and has no q_i as a factor since $\forall i$: $Q \equiv q_i - 1 \pmod{q_i}$.

Conjecture and Proof

Example 6

- We know that some numbers of the form 2^p-1 are prime when p is prime.
  - These are called the Mersenne primes.
- Can we prove the inverse, that a^n-1 is composite whenever either $a>2$, or ($a=2$ but n is composite)?
  - All we need is to find a factor greater than 1.
- Note a^n-1 factors into $(a-1)(a^{n-1}+\ldots+a+1)$.
  - When $a>2$, $(a-1)>1$, and so we have a factor.
  - When n is composite, $\exists r,s>1: n=rs$. Thus, given $a=2$, $a^n = 2^n = 2^{rs} = (2^r)^s$, and since $r>1$, $2^r > 2$ so $2^n - 1 = b^s-1$ with $b = 2^r > 2$, which now fits the first case.

Conjecture & Counterexamples

Example 8

- **Conjecture:** ∀ integers $n>0$, n^2-n+41 is prime.
  - **Hm, let's see if we can find any counter-examples:**
    - $1^2-1+41 = 41$ (prime)
    - $2^2-2+41 = 4-2+41 = 43$ (prime)
    - $3^2-3+41 = 9-3+41 = 47$ (prime) Looking good so far!!
  - Can we conclude after showing that it checks out in, say, 20 or 30 cases, that the conjecture must be true?
- **NEVER NEVER NEVER NEVER NEVER!**
  - Of course, $41^2-41+41$ is divisible by 41!!

Even Great Mathematicians Can Propose False Conjectures!

Example 9

- Euler conjectured that for $n>2$, the sum of $n-1$ n^{th} powers of positive integers is not an nth power.
  - Remained true for all cases checked for 200 years, but no proof was found.
- Finally, in 1966, someone noticed that
  $$27^5 + 84^5 + 110^5 + 133^5 = 144^5.$$
  - Larger counter-examples have also been found for $n=4$, but none for $n>5$ yet.

Fermat's "Last Theorem"

Theorem 2

- **Theorem:** $x^n+y^n=z^n$ has no solutions in integers $xyz \neq 0$ with integer $n>2$.
  - In the 1600s, Fermat famously claimed in a marginal note that he had a "wondrous proof" of the theorem.
    - But unfortunately, if he had one, he never published it!
  - The theorem remained a publicly unproven conjecture for the next ~400 years!
  - Finally, a proof that requires hundreds of pages of advanced mathematics was found by Wiles at Princeton in 1990.
    - It took him 10 years of work to find it!
- **Challenge:** Find a *short, simple* proof of Fermat's last theorem, and you will become instantly famous!

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An Open Problem

- **The 3*x* + 1 Conjecture**: Let T be the transformation that sends an even integer x to $x/2$ and an odd integer x to $3x + 1$. For all positive integers x, when we repeatedly apply the transformation T, we will eventually reach the integer 1.

  For example, starting with $x = 13$:

  $T(13) = 3\cdot 13 + 1 = 40, T(40) = 40/2 = 20, T(20) = 20/2 = 10,$
  $T(10) = 10/2 = 5, T(5) = 3\cdot 5 + 1 = 16, T(16) = 16/2 = 8,$
  $T(8) = 8/2 = 4, T(4) = 4/2 = 2, T(2) = 2/2 = 1$

  The conjecture has been verified using computers up to $5.6 \cdot 10^{13}$.

Some Open Conjectures

- **Conjecture:** There are infinitely many primes of the form n^2+1, where $n \in \mathbf{Z}$.
- **Conjecture:** (Twin Prime Conjecture) There are infinitely pairs of primes of the form $(p, p+2)$.
- **Conjecture:** (The Hailstone Problem) If $h(x) = x/2$ when x is even, and $3x+1$ when x is odd, then $\forall x \in \mathbf{N}\ \exists n \in \mathbf{N}\ h^n(x) = 1$ (where the superscript denotes composition of h with itself n times).

Example 11

Example 12

Example 13

Prove any of these, and you can probably have a lifetime career sitting around doing pure mathematics…

停机问题Halting Problem

- 一个实际问题：
  - 给定一个算法和相应的输入，判定这个算法是否能够完成？（还是进入死循环不能完成？）
  - 由于算法都可以用图灵机描述，所以问题转变为，给定一个图灵机和相应的输入，判定是否停机？
- 对于特定的算法，我们大概是可以判定的，但是否存在对一般的算法进行判定的方法？
  - 程序正确性证明：包括了给一段程序，判断这段程序是否会进入死循环
  - 注意：这个方法本身也是一个算法，否则没有意义

停机问题

- 是否有算法能够判定某个图灵机M在输入I下是否停机？
  - 即Halt(M,I)=True iif M在I处停机，False iif M在I处不停机
  - 答案是否定的，不存在这样的算法
  - 停机问题是不可计算问题

停机问题证明

- 假设存在这样的算法，对应的图灵机是Halt(a,k)，a是要判定的图灵机编码，k是输入的编码
- 我们构造这么一个图灵机Trouble(a)
  - function Trouble(a)
    - if Halt(a, a) = False then
      - return True
    - else
      - loop forever
- Trouble图灵机自然也可以编码，记为t，我们来看把t作为Trouble输入会怎么样？
  - Trouble(t)=?

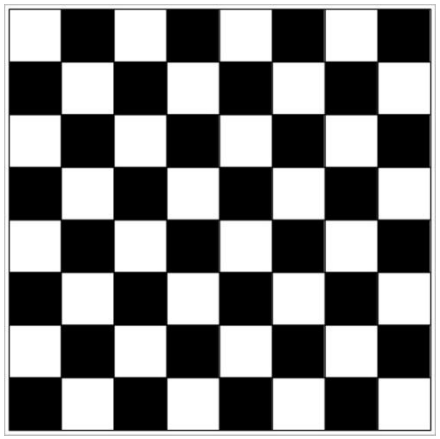
停机问题证明

- 矛盾？矛盾！
  - 如果Trouble(t)停机返回了，也就是Halt(t,t)=False，但这样是说Trouble(t)不停机，矛盾
  - 如果Trouble(t)不停机，也就是Halt(t,t)返回了True，但这样却是说Trouble(t)能停机，矛盾
- 消除矛盾的唯一途径就是否定假设，即不存在能一般性地判定图灵机停机的算法
  - 人的智能就能解决停机问题么？至少目前还是否
  - 对于太长的算法，或者输入太复杂的算法，不能
  - 对于一些短而简单的算法，也不能（一些数论难题）
    - 寻找大于给定N的孪生素数

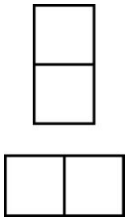
Proof and Disproof: Tilings

Example 1: Can we tile the standard checkerboard using dominos?

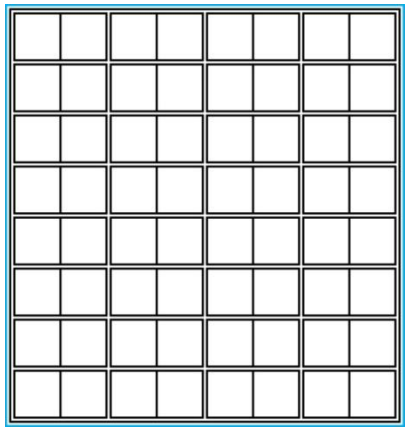
Solution: Yes! One example provides a constructive existence proof.



The Standard Checkerboard



Two Dominoes



One Possible Solution

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Tilings

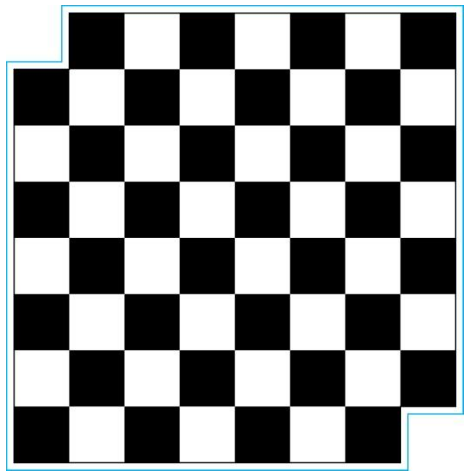
Example 2: Can we tile a checkerboard obtained by removing one of the four corner squares of a standard checkerboard?

Solution:

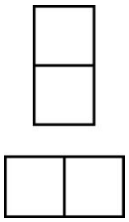
- Our checkerboard has $64 - 1 = 63$ squares.
- Since each domino has two squares, a board with a tiling must have an even number of squares.
- The number 63 is not even.
- We have a contradiction. ◀

Tilings

Example 3: Can we tile a board obtained by removing both the upper left and the lower right squares of a standard checkerboard?



Nonstandard Checkerboard



Dominoes

Continued on next slide →

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Tilings

Solution:

- There are 62 squares in this board.
- To tile it we need 31 dominos.
- *Key fact*: Each domino covers one black and one white square.
- Therefore the tiling covers 31 black squares and 31 white squares.
- Our board has either 30 black squares and 32 white squares or 32 black squares and 30 white squares.
- Contradiction!

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Additional Proof Methods

- Later we will see many other proof methods:
  - Mathematical induction, which is a useful method for proving statements of the form $\forall n\, P(n)$, where the domain consists of all positive integers.
  - Structural induction, which can be used to prove such results about recursively defined sets.
  - Cantor diagonalization is used to prove results about the size of infinite sets.
  - Combinatorial proofs use counting arguments.

homework

- 1.7: 8, 20, 30,
- 1.8: 12, 14, 32, 40